Informal learning in everyday human–LLM interaction

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Abstract

As LLM-based assistants become embedded in work and study, everyday human–LLM interaction is becoming a consequential site of informal learning. Learning science links deeper overt engagement to stronger learning outcomes, making engagement a useful observable process indicator in user-generated interactions when goals are not pre-specified by curricula. Through a large-scale analysis of 128,569 naturalistic conversations, we confirm the presence of learning-oriented engagement at different depths during users' interactions with LLMs: across 491,685 user turns, cognitive engagement appeared in 31.9%, while constructive engagement, the deepest observable form of user engagement, occurred in 4.9%. Our study further reveals the factors associated with these forms of engagement. We found that scaffolded support from the assistant (present in 31.7% of assistant turns) was a consistent factor of richer constructive participation, with associations that varied by user framing, task ecology, support form, timing and prior user state. These findings make behavioural signatures of informal learning observable at scale and broaden how future AI assistance should be evaluated: from the efficiency of answer delivery toward the preservation of cognitive opportunities for users to reason, test ideas and construct understanding in the course of everyday problem-solving.

Keywords: human–AI interaction, informal learning, large language models, scaffolding, learner engagement

1 Introduction

Large language models (LLMs) are becoming embedded in the cognitive work of everyday life. People use them to write, code, search, troubleshoot and make sense of unfamiliar problems, often outside classrooms, training programmes or purpose-built tutoring systems [1–3]. In such moments, the interaction can become more than task completion: assistant responses may create occasions for users to clarify uncertainties, test assumptions, revise ideas and decide how to proceed [4, 5]. Everyday human–LLM interaction therefore constitutes a rapidly expanding setting for informal learning, in which understanding can emerge through self-directed inquiry and problem solving rather than formal curricula [6–8].

Yet existing research on LLMs and learning has focused primarily on settings where learning support is deliberately designed, such as automated feedback, question generation, retrieval practice and tutoring systems [9–13]. These studies show that LLMs can enact instructional strategies when learning is the explicit design target [14, 15]; they tell us less about general-purpose LLM use, where users usually seek task progress rather than instruction. This gap matters because the educational value of everyday AI use depends not only on what models provide, but on what users do with that support: whether an answer becomes material for questioning, testing, revision and extension, or whether it simply accelerates completion while leaving little visible user sense-making [16–20]. Learning science provides a way to make this distinction measurable: deeper forms of overt engagement are associated with stronger learning outcomes, so engagement can serve as an observable process indicator in informal, user-generated interactions whose goals and endpoints are not pre-specified by curricula or study protocols [21, 22]. We therefore ask whether learning-oriented engagement is visible in everyday human–LLM interaction, where its deepest constructive form appears, and how assistant scaffolding is coupled with further constructive participation.

We analyse 128,569 naturalistic conversations from three public large-scale conversational corpora: WildChat, LMSYS Chat and ShareChat [23–25]. Across these corpora, we focus on coding and writing as two common task ecologies in which LLMs are used to produce, revise and troubleshoot knowledge work [1, 4, 26], yielding 491,685 user turns and 489,785 assistant turns. Our analyses move from engagement prevalence, to the contextual organization of constructive engagement, to how assistant scaffolding is associated with user participation at conversation-level, support-form and adjacent-turn scales. To make these questions measurable in naturalistic logs, we translate theories of engagement and scaffolding into role-specific, turn-level behavioural signatures: user turns are coded for passive, active or constructive uptake [21, 22], while assistant turns are coded for scaffolded support and further characterized by support intent and support form [27–30]. We treat constructive engagement as the deepest observable indicator of learning-oriented participation available in these logs, using it to examine how deeper user sense-making is organized by task context, assistant support and local turn ordering.

The results trace a progression from the presence of learning-oriented engagement, to the conditions under which its constructive form appears, to the ways assistant support is coupled with further user participation. First, learning-oriented engagement was recurrent in everyday human–LLM interaction, but its deepest constructive

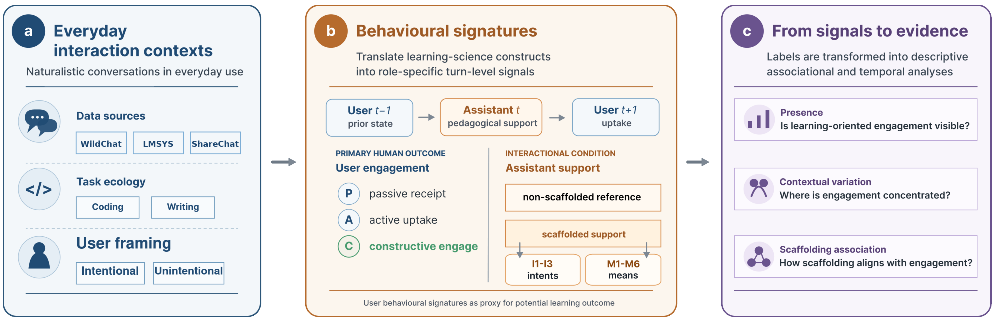

Fig. 1 Analytic framework for studying informal learning in everyday human–LLM interaction. **a,** Public conversations are situated by data source, task ecology and user framing. **b,** Engagement and scaffolding constructs are translated into role-specific turn-level labels for user engagement and assistant support. **c,** These labels support analyses of engagement presence, contextual organization and support–engagement coupling at conversation-level, support-form and adjacent-turn scales. Labels indicate observable learning-oriented participation, not durable learning outcomes or causal instructional effects.

form was selective: cognitive engagement appeared in nearly one third of user turns, whereas constructive engagement appeared in about five percent. Second, constructive engagement was ecologically organized: it was more common under explicit learning-oriented user framing, varied across task ecologies and became most visible in sustained exchanges where users returned to, interrogated and reshaped the assistant's response. Third, assistant scaffolding marked richer constructive participation, but not as a uniform exposure. Its association depended on how support was positioned within the exchange: feedback-like and explanatory forms aligned most clearly with constructive engagement at the conversation level, while adjacent-turn analyses showed that support–engagement coupling in local sequences shaped by prior user state and support form. Together, these findings locate everyday human–LLM interaction in a middle space between ordinary tool use and designed instruction. They suggest that the educational value of future AI assistants lies not in making every interaction tutor-like, but in calibrating when to answer, explain, give feedback or step back, preserving the cognitive responsibility through which people question, test, revise and build judgment while solving real problems.

2 Results

The results follow the analytic sequence in Fig. 1: engagement prevalence, contextual organization and support–engagement coupling. We first quantify cognitive and constructive engagement in everyday human–LLM conversations, then locate constructive engagement across user framing, task ecology and interaction depth. We then turn to assistant support, asking whether scaffolded conversations contain richer constructive participation, which support forms carry this association and how support is ordered with prior and subsequent user engagement at the adjacent-turn scale.

Table 1 Dataset and behavioural overview. Summary statistics for the WildChat, LMSYS Chat and ShareChat task settings analysed in the study. Constructive engagement is reported as the deepest observable level within cognitive engagement and serves as the focal user-side measure in subsequent analyses.

	Sample size			User framing	Assistant support	Cognitive engagement		Emotional engagement
Setting	Conversations	User turns	Assistant turns	Intentional conv. (%)	Scaffolded turns (%)	Overall (%)	Constructive level (%)	Emotional (%)
WildChat coding	31,878	124,073	124,623	37.13	51.03	50.21	9.23	0.95
WildChat writing	39,534	163,705	164,824	25.80	23.53	16.34	2.22	0.57
LMSYS coding	32,114	107,065	105,174	41.49	29.07	45.26	5.44	0.91
LMSYS writing	21,023	77,906	76,279	20.64	19.43	15.30	1.53	0.21
ShareChat coding	2,481	11,541	11,522	45.63	50.67	45.00	15.17	3.38
ShareChat writing	1,539	7,395	7,363	29.69	22.42	33.05	5.18	0.34
Total / pooled	128,569	491,685	489,785	32.12	31.70	31.94	4.93	0.74

Note. User framing is reported as the percentage of conversations with explicit learning-oriented framing. Cognitive, constructive and emotional engagement are percentages of user turns; scaffolded support is computed over assistant turns. The constructive column uses all user turns as the denominator, whereas Fig. 2a decomposes cognitively engaged user turns into passive, active and constructive levels.

2.1 Everyday human–LLM conversations contain learning-oriented engagement

Learning-oriented engagement was visible across all six task settings. Across 491,685 user turns, 31.9% showed cognitive engagement and 4.9% reached constructive engagement, the deepest observable user-side signal in the annotation scheme (Table 1). Visible emotional expression was uncommon, appearing in 0.74% of user turns. Thus, everyday human–LLM conversations often contained observable user uptake of assistant responses, but deeper constructive sense-making was selective rather than routine.

Figure 2a shows how cognitively engaged turns were composed. Active uptake was the dominant form in every setting, accounting for 62.0–86.4% of cognitively engaged turns, whereas constructive engagement accounted for 10.0–33.7% and passive receipt for 1.2–14.6%. This composition separates broad engagement from deeper sense-making. Cognitive engagement shows that everyday LLM use often involves users working with assistant output; active uptake captures use, application or follow-up; and constructive engagement identifies the cases in which users elaborate, test, revise or extend ideas. The remaining analyses therefore focus on where constructive engagement appears and how it is associated with user framing, task ecology, interaction depth and assistant support.

2.2 Constructive engagement is patterned by user framing, task ecology and sustained exchange

Having established that everyday human–LLM conversations contain learning-oriented engagement, we next examined where its constructive form became visible. Explicit learning-oriented framing provided the first contrast (Fig. 2b). In every setting, conversations framed around learning, understanding or exploration had higher cognitive and constructive user-turn rates than conversations without such framing. The cognitive-engagement lift ranged from +17.7 to +47.3 percentage points across settings, and the constructive-engagement lift ranged from +2.5 to +12.8 points. Yet constructive participation was also present in conversations without explicit learning-oriented framing. User framing therefore amplified constructive engagement, but did not define it: learning-relevant participation also emerged in task-oriented exchanges where learning was incidental to solving the problem at hand.

Task ecology provided a second source of variation. Across all three corpora, coding-oriented conversations showed higher cognitive and constructive engagement than writing-oriented conversations. Cognitive engagement ranged from 45.00–50.21% in coding conversations, compared with 15.30–33.05% in writing conversations; constructive engagement ranged from 5.44–15.17% in coding and 1.53–5.18% in writing. A conversation-level context model showed the same direction after including user framing, conversation-length bucket and dataset fixed effects, with coding task ecology positively associated with the presence of at least one constructive user turn (OR=1.89, 95% CI 1.71–2.08, $p < .001$; Supplementary Table C10).

The coding–writing contrast points to task affordances that organize baseline opportunities for visible constructive participation. Coding tasks often externalize uncertainty through errors, constraints and executable tests, creating occasions for users to diagnose failures, refine assumptions, compare alternatives and test candidate solutions. Writing tasks also involve substantial revision and evaluation, but the interaction can often proceed through accepting, adapting or requesting a draft revision without requiring the user to articulate the rationale for the change [31–33]. The observed task difference is therefore consistent with the ecological interpretation: constructive engagement is more likely to appear when the task makes uncertainty, evaluation and revision explicit in the exchange.

The distinction between broad cognitive engagement and constructive engagement reinforces this task-ecology account. LMSYS Chat coding illustrates this distinction: it had high cognitive engagement, yet its cognitively engaged turns were dominated by active uptake rather than constructive elaboration (Fig. 2a). This pattern illustrates that users can work actively with assistant output—applying code, requesting repairs or asking for the next step—without making their reasoning, tests or revisions explicit enough to surface as constructive engagement. Taken together with the coding–writing contrast, this shows that task ecology patterned not only how often users worked with assistant output, but also whether that work became visible as articulated questioning, testing, refinement or extension.

Sustained exchange provided the third context in which constructive participation became visible (Fig. 2c). We operationalized sustained exchange with conversation-length buckets, asking whether a conversation contained at least one constructive user turn. Across all six settings, conversations with more user turns were more likely to contain at least one constructive user turn. The gradient was steeper in coding-oriented settings, but the same direction appeared in writing-oriented settings. This pattern indicates that constructive engagement is most visible when everyday LLM use develops beyond a one-off request for an answer. In such exchanges, users have room to return to the assistant's response, introduce new constraints, test implications, request revisions and reshape their understanding. Because this panel uses an any-event outcome, we treat the length gradient as a visibility pattern; a grouped-binomial

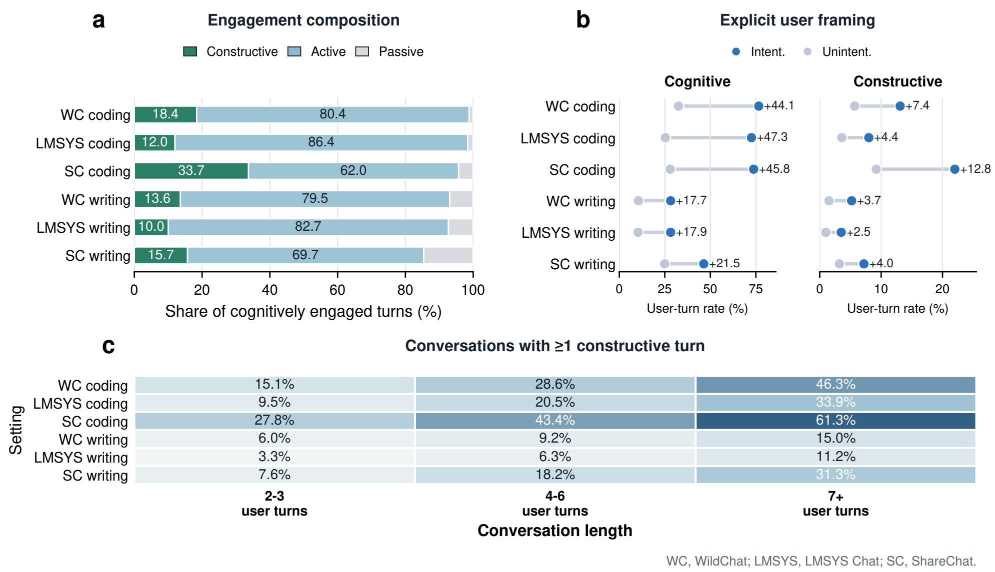

Fig. 2 Learning-oriented engagement varies by engagement form, user framing and conversation length. a, Composition of cognitively engaged user turns by passive, active and constructive levels across the six task settings. **b,** Cognitive and constructive user-turn rates in conversations with versus without explicit learning-oriented framing; annotations show the framed-minus-unframed difference in percentage points. **c,** Share of conversations containing at least one constructive user turn by conversation-length bucket. Panel a uses cognitively engaged user turns as the denominator; panel b reports percentages of user turns; panel c reports conversation-level shares. WC, WildChat; LMSYS, LMSYS Chat; SC, ShareChat.

constructive-rate sensitivity attenuated but preserved the positive length association (4–6 user turns: OR=1.13; 7+ user turns: OR=1.06; Supplementary Table C11).

Together, these results show that constructive engagement was ecologically organized rather than evenly distributed across everyday use. It was amplified by explicit learning-oriented framing, patterned by task affordances that create occasions for testing and revision, and most visible in sustained exchanges. A conversation-level context model with user framing, task ecology, conversation-length bucket and dataset fixed effects supported the same organization, and the rate sensitivity showed that the length association was attenuated but preserved after accounting for user-turn denominators (Supplementary Tables C10 and C11). Having located these user-side and contextual patterns, we next ask whether assistant scaffolding marks conversations with richer constructive participation.

2.3 Scaffolded support marks richer constructive participation

Having located the user-side and contextual conditions under which constructive engagement became visible, we next turned to assistant-side support. At the conversation level, conversations containing at least one scaffolded assistant turn had higher

Table 2 Summary of key support–engagement associations across the three corpora. Constructive-ratio contrasts compare conversations with versus without scaffolded support using turn-weighted constructive user-turn ratios. Adjusted models report the association of scaffolded-support presence with constructive-turn count (Poisson count ratio) and with having at least one constructive turn (logit odds ratio). Adjacent-turn lift is the difference in the probability that the next user turn is constructive after scaffolded versus non-scaffolded reference assistant turns.

Setting	Constructive ratio Has Scaf. / No Scaf.	Difference (pp)	Depth diff. (turns)	Poisson count ratio	Logit OR	Adjacent lift (pp)
WildChat coding	0.101 / 0.056	+4.6	+2.23	1.852	1.761	+2.68
LMSYS coding	0.065 / 0.040	+2.5	+1.42	1.622	1.544	+2.01
ShareChat coding	0.163 / 0.090	+7.3	+3.36	1.893	2.140	+6.21
WildChat writing	0.027 / 0.017	+1.0	+3.13	1.569	1.437	+1.13
LMSYS writing	0.022 / 0.011	+1.1	+2.10	1.985	1.764	+0.27
ShareChat writing	0.066 / 0.034	+3.2	+3.38	2.491	1.757	+3.35

turn-weighted constructive user-turn ratios than reference conversations without scaffolded support in all six task settings (Fig. 3a). The constructive-ratio differences ranged from +0.99 to +7.34 percentage points across WildChat, LMSYS Chat and ShareChat, and all six contrasts were distinguished from zero in conversation-level bootstrap tests ($p < .001$; Supplementary Table C8). Scaffolded conversations also showed more turns after the first assistant response in every setting (+1.42 to +3.38 turns; Fig. 3d), indicating that scaffolded support marked exchanges with more sustained participation. Together, these descriptive contrasts show that scaffolded support co-occurred with richer constructive participation across corpora and task settings.

The association remained positive after accounting for observed user framing and interactional context in covariate-adjusted models. In setting-specific models, scaffolded-support presence was associated with higher constructive-turn counts, with Poisson count ratios ranging from 1.569 to 2.491, and with higher odds that a conversation contained at least one constructive user turn, with logistic odds ratios ranging from 1.437 to 2.140 (Fig. 3b). The same positive direction appeared within both explicitly learning-oriented and non-explicitly framed conversations (Fig. 3c), indicating that the support–engagement association was not limited to interactions users announced as learning episodes. A rate sensitivity treating user turns as the exposure preserved positive scaffolded-support associations in all six settings (rate ratios 1.356–1.703; all $p < .001$; Supplementary Table C9). Across these checks, scaffolded support remained a consistent conversation-level factor of richer constructive participation. This conversation-level signal motivates a finer analysis of the support forms through which scaffolded support was expressed.

2.4 Support form differentiates the scaffolding–engagement association

The conversation-level analyses established scaffolded support as a consistent factor of richer constructive participation. We next asked which features of scaffolded support carried this signal. For each scaffolded assistant turn, the annotation scheme recorded two parallel descriptors: a support-intent label indicating whether the turn primarily targeted metacognitive, cognitive or affective support (I1–I3), and one or

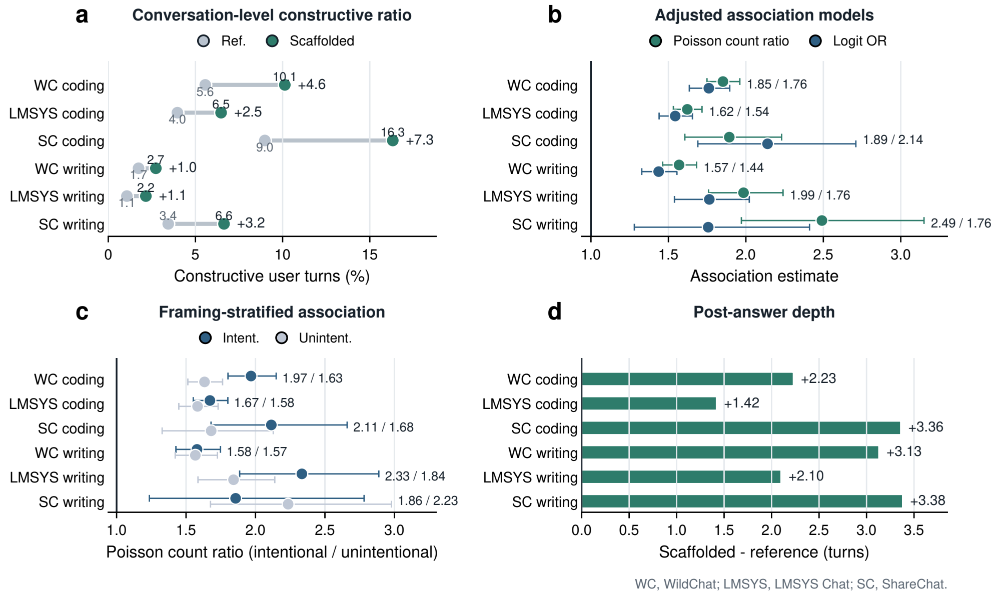

Fig. 3 Scaffolded support marks richer constructive participation across task settings. **a,** Turn-weighted constructive user-turn ratios in conversations containing scaffolded support versus reference conversations without scaffolded support. **b,** Covariate-adjusted within-setting associations between scaffolded-support presence and constructive participation, reported as Poisson count ratios and logistic odds ratios. **c,** Poisson count ratios for scaffolded-support presence stratified by explicit versus non-explicit learning-oriented framing. **d,** Difference in the number of turns after the first assistant response between scaffolded and reference conversations. Numbers indicate scaffolded-minus-reference differences, except panels b and c, which report model estimates. Error bars in panels b and c show model-based 95% confidence intervals; bootstrap confidence intervals for the descriptive contrasts in panels a and d are reported in Supplementary Table C8. WC, WildChat; LMSYS, LMSYS Chat; SC, ShareChat.

more support-form labels indicating how the support was delivered—feedback, hinting, instructing, explaining, modelling or questioning (M1–M6). This two-axis scheme follows scaffolding accounts that distinguish the purpose of assistance from the means through which assistance is provided [27–30]. Support intent offered limited contrast because scaffolded turns were predominantly cognitive in purpose (Supplementary Table C16). We therefore focused on the form axis, comparing scaffolded conversations containing each support form with scaffolded conversations without that form and reporting turn-weighted constructive user-turn ratio differences (Fig. 4a,b).

Support form sharply differentiated the scaffolded-support association. Feedback-like support (M1) showed the largest constructive-engagement contrast in both explicitly framed and non-explicitly framed conversations, with the association strongest under explicit learning-oriented framing (+14.7 versus +6.7 percentage points; framing difference, +8.0 points; Fig. 4a). Explaining (M4) showed the same direction (+6.7 versus +3.0 points; framing difference, +3.6 points), and hinting (M2) was positive but smaller. Instructing (M3) and questioning (M6), by contrast, were lower or negative, especially in explicitly framed conversations. Task-stratified estimates gave a complementary view (Fig. 4b): feedback-like support (M1) and explaining (M4) were positive in both coding- and writing-oriented task ecologies, whereas instructing (M3), modelling (M5) and questioning (M6) varied more strongly by task.

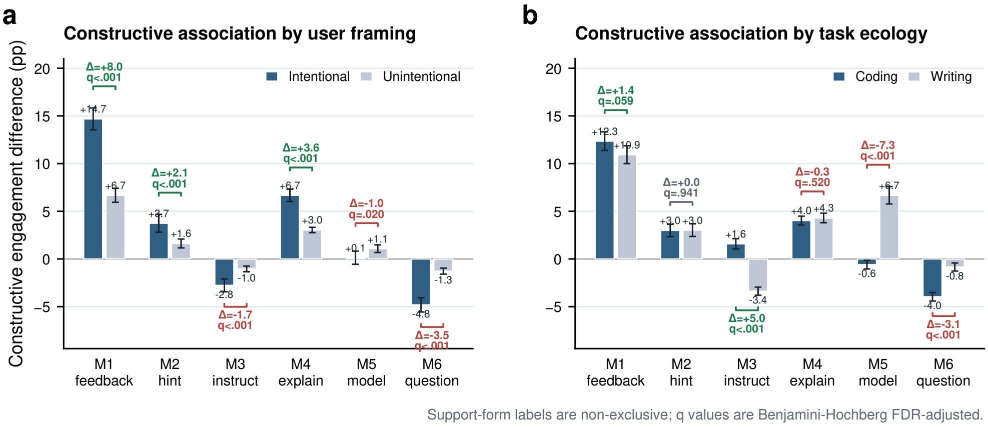

Fig. 4 Support forms differentiate scaffolded interaction. a, Constructive-engagement differences associated with each support form among scaffolded conversations, stratified by intentional versus unintentional user framing. **b,** The same support-form contrasts stratified by coding- versus writing-oriented task ecology. Bars are turn-weighted constructive user-turn ratio differences, comparing scaffolded conversations containing a support form with scaffolded conversations without that form; error bars show 95% confidence intervals, and bracket annotations show between-stratum differences and Benjamini–Hochberg FDR-adjusted q values within each panel family. Support-form labels correspond to M1–M6 and are not mutually exclusive.

The form-level pattern is distinctive because these support moves occurred inside general-purpose task assistance rather than within a designed lesson. In formal instruction, feedback, explanation and direct guidance are usually embedded in planned sequences with explicit learning goals. In everyday LLM use, the same forms are interleaved with task completion, so an assistant response can either close the problem by supplying the next step or leave part of the problem available for the user to inspect, revise or test. Feedback-like support (M1) was strongly aligned with constructive engagement because it takes the user's prior attempt as the object of the response, making a mismatch, partial success or error available for evaluation, repair or extension [34–36]. Explaining (M4) similarly opens the assistant's answer by exposing mechanisms or rationales that users can test, adapt or connect to their own understanding [37, 38]. Hinting (M2) can leave solution work open, but its association was smaller. Instructing (M3), modelling (M5) and questioning (M6) can be useful for task progress, examples or clarification; in these logs, however, they were less consistently aligned with constructive elaboration. This is consistent with assistance-dilemma and scaffolding accounts in which highly executable steps, models to imitate or information-management moves can reduce the need for learners to generate, evaluate or revise the next idea [20, 39, 40]. Thus, in everyday human–LLM interaction, support form is informative not only as a pedagogical category but as an interactional position: it indicates whether assistance turns the answer into material for the user's next cognitive move or completes the deliverable with little further reasoning visible.

The support-form supply profile adds task context to these associations (Supplementary Fig. D3). Explaining (M4) dominated scaffolded coding turns across the three corpora, whereas writing-oriented support drew more evenly on instructing (M3), explaining (M4) and questioning (M6). This support ecology clarifies the interactional environment in which the form-level associations arose: coding and writing differed both in which support forms aligned with constructive engagement and in which scaffolding repertoires users typically encountered.

Together, these analyses refine the conversation-level support result. Scaffolded support marked richer constructive participation, and this association was carried most clearly by particular support forms within task-specific scaffolding repertoires. The form-level result makes the temporal question sharper: if support matters partly through how it responds to the user's ongoing work, then its adjacent-turn association with constructive engagement should depend on the user's immediately preceding state. We therefore next examine support–engagement coupling at the adjacent-turn scale.

2.5 Support–engagement coupling is state- and form-dependent

The preceding analyses established two things: scaffolded conversations contained richer constructive participation, and this association was carried unevenly by support form. These results identify where assistant support is associated with constructive engagement, but not the interactional scale at which the association appears. We therefore turned to adjacent-turn ordering, analysing how the user's current state, the assistant's support form and the immediately following user turn were linked.

At this adjacent-turn scale, scaffolded assistant turns were more likely than reference assistant turns to be followed by constructive engagement in all six task settings (Fig. 5a). The next-turn constructive lift was positive in every setting, ranging from +0.27 to +6.21 percentage points. Five of the six adjacent-turn lifts were statistically distinguishable from zero in conversation-cluster bootstrap tests; LMSYS Chat writing was positive but weaker (+0.27 percentage points, 95% CI -0.01 to +0.56, p = .061; Supplementary Table C8). The temporal signal was therefore consistently positive across settings, but heterogeneous in magnitude.

This adjacent-turn coupling was organized by the user's immediately preceding state. State-conditioned contrasts were generally positive but varied across prior constructive, active and passive turns (Fig. 5b). The user-to-assistant direction showed the same contingency: constructive and active user turns were more often followed by scaffolded support than passive turns, especially in coding-oriented conversations (Fig. 5c). These two directions suggest a coupled sequence rather than a one-way support pattern: engaged user states more often preceded scaffolded support, and scaffolded support was then more often followed by further constructive participation.

We then asked whether the form of support shaped this adjacent-turn coupling once prior user state and adjacent-turn context were modelled jointly. We fitted integrated logistic regressions predicting whether the next user turn was constructive,

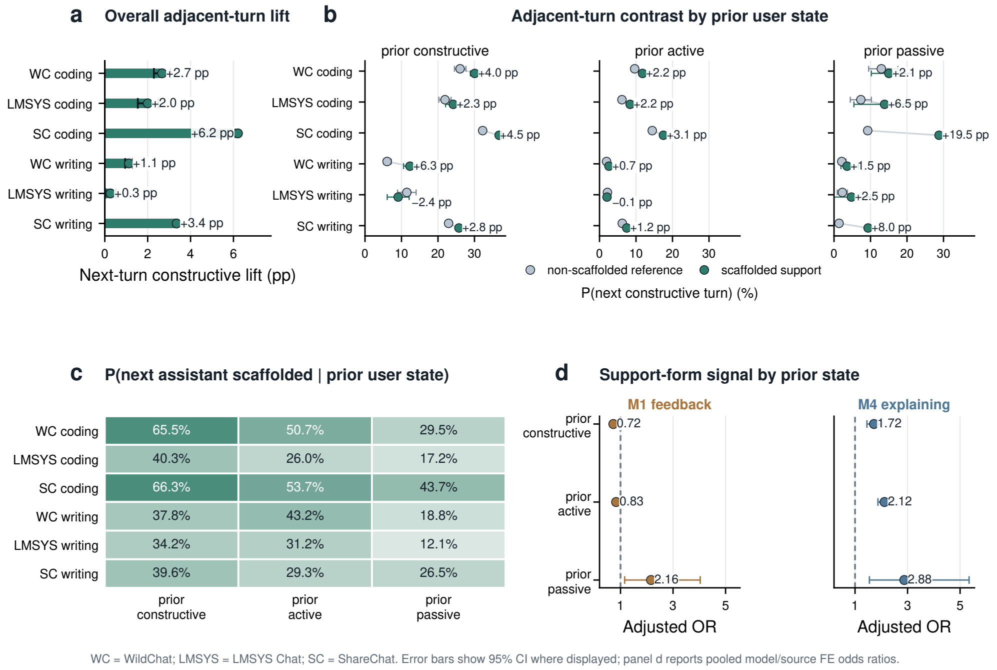

Fig. 5 Support–engagement coupling is state- and form-dependent across task settings. **a,** Next-turn constructive lift after scaffolded versus reference assistant turns. **b,** Probability of a constructive next user turn by prior user state and assistant support condition; annotations show scaffolded-minus-reference differences. **c,** Probability that the next assistant turn contains scaffolded support by prior user state. **d,** Pooled model/source fixed-effect odds ratios for focal support forms within prior user states. M1 feedback and M4 explaining are shown because Section 2.4 identified them as focal positive conversation-level forms; full M1–M6 interaction estimates and representative support-form cards are reported in Supplementary Table C15 and Supplementary Fig. D4. Error bars show 95% confidence intervals where displayed. Panel d reports adjacent-turn logistic models with conversation-cluster robust standard errors. WC, WildChat; LMSYS, LMSYS Chat; SC, ShareChat.

combining prior user state, assistant-scaffolding descriptors, task and framing variables, turn position and model/source fixed effects. The scaffolding block improved fit beyond user state and context (full scaffolding block LR $\chi^2_7 = 1339.6$, $p < .001$; Supplementary Table C14), and prior-state × support-form interactions improved fit further (pooled model/source fixed effects: LR $\chi^2_{18} = 297.3$, $p < .001$; Supplementary Table C15). These tests show that the immediate support–engagement association was not reducible to the user's preceding state alone; it depended jointly on what the user had just made visible and how the assistant supported the next step.

Figure 5d highlights feedback-like support and explaining, the two support forms that carried the clearest positive signal in the conversation-level analysis; complete M1–M6 interaction estimates and paraphrased support-form cards are reported in the Appendix (Supplementary Table C15; Supplementary Fig. D4). Explaining provided the more stable adjacent-turn signal, predicting constructive follow-up after prior constructive, active and passive user turns (ORs 1.72, 2.12 and 2.88, respectively; all

$p < .001$). Feedback showed a different immediate role: its positive next-turn association was concentrated after prior passive turns, rather than after turns in which users were already active or constructive. This divergence clarifies the conversation-level result: feedback marked exchanges in which user attempts were available for evaluation and repair, whereas explanation more consistently provided material for the user's immediate next move. Support form therefore mattered through its position in the unfolding user–assistant sequence, not as a context-free pedagogical category.

Taken together, the temporal analyses locate support–engagement coupling in adjacent-turn sequences. Constructive participation was most clearly organized where the user's visible state, the assistant's support condition and the form of support were considered together. Learning-oriented engagement in everyday human–LLM interaction is therefore assembled in situated next-turn sequences in which the user's visible work and the assistant's support form jointly shape whether constructive participation continues.

3 Discussion

Everyday human–LLM interaction as an informal learning ecology

By tracing engagement in large-scale public logs, this study shows that everyday human–LLM interaction can be analysed as an empirical ecology of informal learning. These conversations are organized around users' own tasks, yet they contain behavioural traces of learning-oriented work. Section 2.1 establishes the depth structure of this participation: cognitive engagement appeared in nearly one third of user turns, whereas constructive engagement appeared in about five percent. This distinction matters because broad uptake and constructive sense-making mark different relationships to AI output. Uptake shows that users incorporate, apply or extend an assistant response in the task at hand; constructive engagement marks the more selective moments in which users make their reasoning visible by articulating constraints, testing implications, revising assumptions or extending explanations. In this way, informal learning becomes observable not as a declared intention, but as a pattern of participation within ordinary interaction.

The contextual results in Section 2.2 show how this participation was organized. Explicit learning-oriented framing made constructive engagement more common, but task-oriented exchanges also contained constructive turns. Task ecology shaped which forms of sense-making became visible: coding conversations often exposed failure, constraints and tests, whereas writing conversations made revision, adaptation and evaluative choice visible when users brought those concerns into the exchange. Sustained conversations added temporal opportunity, allowing earlier responses to become points of return, comparison and revision. Together, these findings define everyday human–LLM interaction as an informal learning ecology: user orientation, task affordances and interaction depth shape when ordinary problem solving becomes an occasion to build, test and extend understanding.

Scaffolding as situated support

The assistant-side findings show how support enters this ecology. As shown in Section 2.3, conversations containing scaffolded support had richer constructive participation across all six task settings, and this association remained positive after adjustment for observed conversation-level context. This pattern is most informative when scaffolded support is read as a relation between the assistant response and the user's next possible action. In everyday task-oriented LLM use, an answer can do more than provide a usable deliverable: it can expose a rationale, error, mechanism, constraint or partial solution that remains available for inspection. This view extends scaffolding theory, which emphasizes contingent assistance and the gradual transfer of responsibility [27, 28, 40], into general-purpose AI interaction, where task completion and opportunities for sense-making are interleaved within the same exchange.

The support-form results specify what this situated support looked like in the logs. As shown in Section 2.4, feedback-like support and explanation were most consistently aligned with constructive engagement. Feedback makes a prior attempt evaluable: it turns a user's partial solution, diagnosis or draft into something that can be repaired, refined or extended [34, 35]. Explanation makes reasons and mechanisms inspectable, giving users material for comparison, self-explanation and transfer [37, 38]. Other forms, including instruction, modelling and questioning, supported task progress, examples and clarification, with more task-contingent alignment to visible constructive elaboration. The form-level contrasts point to the warrant that support attaches to generated content: some forms organize action, whereas others make visible the grounds on which an answer, attempt or mechanism can be used. For general-purpose AI assistance, this shifts scaffolding from the surface form of a response to the epistemic status of the output it produces: fluent answers become learning-relevant when they are also judgeable answers, with grounds, limits or failure points that can be inspected and adapted within the task.

Learning continuity across formal and everyday contexts

The temporal findings clarify what everyday AI assistance can add to a broader learning system. As shown in Section 2.5, support–engagement coupling was most interpretable in adjacent user–assistant–user sequences, where the user's preceding state and the assistant's support form shaped whether constructive participation continued. This scale gives everyday AI use a role that differs from formal instruction. Formal learning environments organize progression through curricula, sequencing, assessment, shared standards and opportunities to abstract beyond the immediate task. Everyday AI assistance organizes re-entry: it brings explanations, feedback and prior reasoning back into the moments where people apply knowledge under practical constraints.

This distinction matters because much learning depends on whether knowledge can be reactivated when it is needed. A concept explained in a course, tutorial or earlier exchange becomes educationally consequential when it can be recognized again in a new problem, adapted to a different constraint or used to diagnose a breakdown.

Human–LLM exchanges provide occasions for that re-entry. A user encounters a concrete difficulty; the assistant supplies a rationale, comparison or correction; the user can then relate that material to the task at hand. The value of such episodes lies in making knowledge usable within action, where understanding is tested against the demands of a real problem.

This complementarity suggests a different role for personal AI assistants in education. Schools, courses and curricula provide progression, accountability and shared criteria. AI assistants can provide continuity across contexts: carrying explanations from one task to another, preserving traces of prior attempts, helping users notice recurring uncertainties and supporting consolidation after repeated encounters with a concept. In this role, lifelong AI assistance can function as a connective layer between structured learning and everyday practice, giving memory and personalization to the informal learning that emerges while people work.

AI-mediated work and the social organization of expertise

The broader implication is that AI-mediated work may reshape the pathways through which expertise is formed. Much professional learning develops through participation in the practices of a domain: proposing first attempts, encountering error, receiving feedback, revising choices, explaining decisions and seeing consequences unfold [41, 42]. These activities are often treated as parts of production, yet they also carry an apprenticeship function. The results provide an empirical entry point into this process in AI-mediated settings: learning-oriented participation was measurable, selective and structured by the conditions of interaction. As LLMs become embedded in writing, coding, search and problem solving, the formation of expertise becomes tied to how these environments organize participation in the practices through which judgment is built.

AI also changes the repertoire of capacities through which people enter expert practice. When systems can generate fluent drafts, code, explanations and recommendations, expertise may depend increasingly on formulating good problems, recognizing relevant principles, interrogating generated solutions, judging evidence, revising under constraints and remaining accountable for consequences. These capacities extend beyond generic AI literacy; they are forms of situated judgment that develop through repeated participation in domain work. Schools, workplaces and platforms will therefore shape future expertise through the routines they build around AI: what novices practise, where feedback enters, how responsibility is assigned and how mistakes become occasions for development.

A science of everyday AI learning should therefore study cognitive apprenticeship as a distributed property of sociotechnical systems. The outcomes of interest extend beyond immediate productivity or single-session engagement to the formation of future practitioners: whether repeated AI-mediated routines cultivate judgment, autonomy and domain competence over time. The social stakes are distributional. Intelligent assistance can broaden access to expert-like feedback, explanation and revision, but the developmental value of that access will depend on institutional design: who participates in diagnosis, who practises revision, who observes consequences and who is asked to justify decisions. The long-term educational significance of AI will depend on how societies organize these pathways into expertise, and on whether AI-mediated work sustains broad participation in the practices through which judgment, autonomy and domain competence are formed.

Limitations

Several boundaries define the interpretation of these findings. The analyses use public conversational logs and characterize population-level interactional patterns rather than longitudinal learner histories. They do not observe users' prior knowledge, motivation, goals outside the conversation or later learning outcomes. The engagement labels capture behavioural manifestations of learning-oriented participation, not direct evidence of retention, transfer or skill development. Assistant support was also observed rather than assigned, so the associations reported here should be interpreted as structured patterns in naturalistic interaction rather than causal effects of scaffolding. Finally, the analysis focuses on English-language coding and writing conversations from three public corpora; other domains, languages, interfaces and institutional settings may organize different opportunities for engagement and support. These boundaries motivate the next stage of evidence: connecting the behavioural signatures identified here to independent measures of learning, autonomy and transfer in controlled and longitudinal settings.

Conclusion

As LLMs become infrastructure for everyday work and study, the central educational question extends beyond how individuals learn from intelligent systems to how learning opportunities are organized in AI-mediated societies. This study shows that ordinary human–LLM interaction already contains measurable behavioural signatures of informal learning, and that deeper constructive engagement emerges selectively through task ecologies, support forms and short interactional sequences. The long-term significance of AI may therefore depend not only on the knowledge or productivity it delivers, but also on how societies organize learning opportunities: who encounters them, how often they arise and whether institutions sustain them over time.

4 Methods

4.1 Study design

This study is an observational analysis of naturalistic public human–LLM conversations from WildChat-4.8M [23] (reported below as WildChat) , LMSYS Chat-1M [24] (reported below as LMSYS Chat) and the ShareChat [25] strict-English subset (reported below as ShareChat) across two everyday task ecologies, coding and writing. It asks whether learning-oriented engagement is visible in ordinary use, where constructive engagement is concentrated and how assistant scaffolding is ordered with user engagement across conversation-level and adjacent-turn timescales. The design is descriptive and associational, following the logic of large-scale digital trace and computational social-science analyses [43, 44]. It does not estimate randomized exposure effects, causal instructional efficacy or durable learning outcomes [45].

The empirical sequence mirrors the Results section. We first estimate the prevalence and composition of user engagement, then examine how constructive engagement varies by user framing, task ecology and interaction depth. We next test whether scaffolded assistant support marks conversations with richer constructive participation, separate scaffolded support by support form and analyse adjacent-turn coupling between assistant support and user engagement. Dataset and task ecology are treated as substantive dimensions of the interaction environment rather than nuisance variables to be averaged away.

4.2 Data sources and analytic samples

The analytic corpus contains 128,569 conversations and 981,470 total turns, including 491,685 user turns and 489,785 assistant turns. The six task settings are WildChat coding, WildChat writing, LMSYS Chat coding, LMSYS Chat writing, ShareChat coding and ShareChat writing. WildChat and LMSYS Chat provide large public human–LLM interaction logs from deployed chat systems, whereas ShareChat provides a smaller publicly shared conversation corpus after English-language filtering; together they offer complementary naturalistic settings rather than experimentally matched platform conditions [23–25]. WildChat coding contributes 31,878 conversations with 124,073 user turns and 124,623 assistant turns. WildChat writing contributes 39,534 conversations with 163,705 user turns and 164,824 assistant turns. LMSYS Chat coding contributes 32,114 conversations with 107,065 user turns and 105,174 assistant turns. LMSYS Chat writing contributes 21,023 conversations with 77,906 user turns and 76,279 assistant turns. ShareChat contributes 2,481 coding conversations with 11,541 user turns and 11,522 assistant turns and 1,539 writing conversations with 7,395 user turns and 7,363 assistant turns.

The six task settings were analysed with a common Level 1–4 annotation and analysis framework so that prevalence, contextual variation, support–engagement association and temporal ordering were computed from the same construct definitions. The manuscript reports the refreshed production outputs after the latest user-turn update pass and the subsequent LMSYS Chat and ShareChat annotation runs rather than earlier pre-update batches. Supplementary Section A.1 describes the corpus construction and refresh provenance in more detail.

Corpus filtering and classification are reported in Supplementary Table A1, with retained provenance artifacts and script-hash prefixes in Supplementary Table A2. In brief, coding and writing task ecologies were assigned by English-oriented LLM-assisted semantic filters rather than by native topic metadata alone. ShareChat additionally underwent a strict-English screen after task filtering, retaining only conversations whose final conversation-level language flag was English. For ShareChat, public dataset-card turns, raw CSV message rows and role-normalized analysis turns are distinct counting units; the conversion therefore uses platform-url conversations and role-labelled message rows before deriving user and assistant turns. Supplementary Table A3 summarizes the available validation and audit evidence for the task, language and user-framing classifiers.

4.3 Conversation metadata and task ecologies

Each conversation is represented as an ordered sequence of user and assistant turns together with available metadata. The main contextual variables are data source, task domain, user framing, topic and language. Task domain and user framing are analytic variables generated by the filtering and annotation pipeline; topic and language are retained as corpus or pipeline metadata where available. Coding and writing were selected because they are common everyday LLM use cases but differ in their interactional affordances: coding often involves debugging, error diagnosis and iterative testing, whereas writing often involves drafting, revision and directive editing [31–33]. User framing is a conversation-level LLM-assisted label that distinguishes conversations explicitly oriented toward understanding, exploration or learning from conversations without such explicit learning-oriented framing.

Behavioural depth is measured from the number and ordering of turns in the conversation. Depth is used descriptively, as a factor of interactional opportunity and sustained exchange, not as an independent manipulation. Post-answer depth is measured as the number of turns after the first assistant response.

4.4 Turn-level constructs

The unit of annotation is the conversational turn. User and assistant turns are labelled with different constructs because they play different roles in the interaction. User turns are labelled for behavioural, emotional and cognitive engagement, drawing on engagement theory and the ICAP distinction between passive, active and constructive participation [21, 22]. The main analyses emphasize cognitive engagement and its depth levels. Passive engagement (`P`) captures reception or acknowledgement without substantial transformation. Active engagement (`A`) captures task-relevant action, follow-up or manipulation of supplied information. Constructive engagement (`C`) captures elaboration, revision, testing, extension or transformation of ideas. Constructive engagement is treated as the highest-level behavioural proxy for learning-oriented participation available in these logs, not as a direct measure of knowledge gain.

Assistant turns are labelled for scaffolding. Scaffolded support denotes assistant behaviour that structures, regulates or redirects the user's process while leaving some cognitive work with the user; the comparison category is a non-scaffolded reference response, including straightforward task fulfilment or stand-alone information provision. This definition follows the scaffolding tradition, especially contingency, transfer of responsibility and potential fading [27, 28, 30]. Scaffolded turns can also receive intention labels (`I1` metacognitive, `I2` cognitive and `I3` affective support) and non-mutually exclusive means labels (`M1` feedback, `M2` hinting, `M3` instructing, `M4` explaining, `M5` modelling and `M6` questioning). Operational examples are provided in Supplementary Table C1.

4.5 Annotation workflow and verification

The annotation workflow combined theory-grounded codebook development, prompt iteration, parser checks, human review and production-scale LLM-assisted labelling.

We used LLM-assisted annotation as a scalable text-analysis workflow while retaining parser checks and human verification because prior work shows both the promise and the need for validation when language models are used as annotators [46, 47]. Prompt and post-processing rules were first developed on validation batches to reduce common failure modes, including over-labelling brief acknowledgements as active engagement and misclassifying directive assistant responses as scaffolding. After the pipeline stabilized, the same annotation logic was applied to the WildChat, LMSYS Chat and ShareChat analytic corpora. The archived annotation metadata record the provider, deployment name, API version, prompt version, retry policy, parser checks and rule-based post-processing rules used for production labelling; these details are summarized in Supplementary Section A.3. All examples reproduced in the manuscript are de-identified and paraphrased rather than copied from raw logs.

Human verification was used as a quality-control input rather than as a separate source of outcome measurement. We distinguish three validation roles. First, human–human agreement assesses whether reviewers could apply the codebook consistently. Second, production-label checks compare final or post-processed production labels with human review where a direct audit was available. Third, post-update audits check whether targeted update passes reduced known residual false positives. The final coding validation rerun combines three review rounds, with 658 task turns and 643 turns after deduplication. The final writing validation used 210 reviewed conversations. These validation sets were organized by task ecology rather than independently powered within each public corpus. Consolidated human-confirmed production-label audits covered user framing, 600 user turns and 180 assistant turns across the six corpus-by-task settings; these audits indicated acceptable or better reliability across user engagement, user framing, assistant scaffolding and support-form labels. WildChat also underwent a final targeted user-turn update check; LMSYS Chat and ShareChat were then processed with the same codebook, parser checks and post-processing logic. Agreement was summarized with per-label F1, MCC and Gwet's AC1 because the labels are sparse, prevalence-imbalanced and not mutually exclusive in all cases [48, 49]. All reported writing labels exceeded F1=0.70 in the human–human codebook check. The primary claims rely most heavily on constructive engagement, scaffolded support and support-form labels. Supplementary Tables C2 and C3 report human–human codebook agreement and consolidated human–LLM production-label agreement, and Supplementary Section A.4 describes the final user-turn update check, the user-framing audit and corpus-transfer scope.

4.6 Outcome construction

The primary user-side outcome is constructive engagement. At the turn level, this is a binary indicator for whether the user turn is labelled `C`. At the conversation level, we use both the count of constructive user turns and an indicator for whether a conversation contains at least one constructive user turn. Constructive ratios are defined as constructive user turns divided by user turns.

The primary assistant-side exposure is scaffolding, operationalized as scaffolded support. At the turn level, this is the distinction between scaffolded support and a non-scaffolded reference response. At the conversation level, we use an indicator for whether a conversation contains at least one scaffolded assistant turn. Support-form analyses condition on scaffolded turns and compare conversations containing each means label with scaffolded conversations that do not contain that label. Because means labels can co-occur, these are descriptive support-form contrasts rather than mutually exclusive treatment comparisons.

4.7 Statistical analyses

Descriptive analyses estimate engagement prevalence, scaffolding prevalence, passive/active/constructive composition, emotional-engagement rates and behavioural-depth summaries within each task setting. Context analyses compare intentional versus unintentional conversations and coding versus writing settings. Depth analyses group conversations into behavioural-depth buckets and estimate the probability that a conversation contains at least one constructive turn. Because any-event outcomes mechanically become easier to observe in longer conversations, we also fit a grouped-binomial constructive-rate sensitivity model in which the outcome is the constructive user-turn count out of total user turns, with user framing, task ecology, length bucket and dataset fixed effects as predictors.

Conversation-level associations are analysed by comparing conversations with and without at least one scaffolded assistant turn. Constructive-ratio contrasts use turn-weighted ratios within each comparison group, defined as total constructive user turns divided by total user turns, with uncertainty from 1,000 conversation-level bootstrap resamples. The adjusted Section 2.3 count model is a setting-specific Poisson generalized linear model for the raw constructive-turn count per conversation, not for a rate. For each setting s, the generic mean model was $\mathbb{E}(C_i) = \exp(\alpha_s + \beta_s S_i + \gamma_s^\top X_i)$, where C_i is the constructive-turn count, S_i indicates at least one scaffolded assistant turn and X_i contains observed conversation-level context such as user framing, conversation length, emotional-expression ratio and error or persistence markers where available. We did not use a user-turn offset in the primary count model; conversation length entered directly as a covariate. The exponentiated coefficient for scaffolded-support presence is reported as a multiplicative association with expected constructive-turn count conditional on observed covariates. As a rate sensitivity, we refitted the same setting-specific mean model with user turns represented as an exposure offset and quasi-Poisson scaled standard errors.

The companion binary model for Section 2.3 is a setting-specific logistic regression for whether a conversation contains at least one constructive user turn, using scaffolded-support presence and the same core conversation-level context available within settings. Task ecology is fixed within each setting and is therefore used in pooled/context models, not interpreted as a within-setting coefficient. Supplementary Table C6 lists the estimable covariate sets used for the setting-specific count, binary and offset-rate models. Model-based standard errors and 95% confidence intervals are shown for the adjusted Poisson and logistic estimates. We checked Poisson overdispersion using the Pearson χ^2 statistic divided by residual degrees of freedom; the dispersion estimates ranged from 1.16 to 2.06 across the six settings. A quasi-Poisson sensitivity that scaled the Poisson standard errors by this dispersion preserved the direction and statistical distinguishability of the scaffolded-support association in every setting, and the offset-rate sensitivity likewise preserved positive scaffolded-support associations in every setting. Negative-binomial models were not used as the primary specification because the estimand was the common pre-specified multiplicative association under the same mean model across settings; the overdispersion and offset checks were used to assess variance and exposure-specification sensitivity rather than to change the main estimand. Binary outcomes in other analyses, including whether the next user turn is constructive and whether the next assistant turn is scaffolded, are modelled with logistic regression or reported as conditional probability contrasts, depending on the estimand [50, 51]. Percentage-point contrasts are reported as effect sizes; adjacent-turn contrasts use 1,000 conversation-cluster bootstrap resamples because multiple turn pairs can come from the same conversation. Support-form bracket tests in Fig. 4 use Benjamini–Hochberg false-discovery-rate adjustment within each displayed family of six support-form comparisons. Supplementary Tables C4–C7 summarize the main model families and temporal estimands.

To assess whether the adjacent-turn pattern was reducible to a single feature family, we also fitted integrated logistic regressions that combined user-state features, assistant-scaffolding features and adjacent-turn context in one model. The outcome was whether the next user turn was constructive. The pooled model included dataset fixed effects; a sensitivity replaced these with recoverable assistant model or source-family fixed effects, using exact `chat_model` labels where available, LMSYS Chat model names recovered from conversation identifiers and ShareChat assistant/source families recovered from conversation identifiers. Standard errors for these adjacent-turn regressions were clustered by conversation. We report broad scaffolded-support models separately from support-form decompositions because M1–M6 are nested descriptors of scaffolded support rather than independent exposures. Nested likelihood-ratio block tests first add broad scaffolded-support presence and then test whether support-form descriptors add signal within scaffolded support. These model/source checks are interpreted as robustness analyses rather than model-causal comparisons.

Temporal analyses focus on adjacent-turn ordering. The primary analyses compare the probability that the next user turn is constructive after scaffolded versus non-scaffolded assistant turns, estimate the same contrast within prior user states and estimate the probability that the next assistant turn is scaffolded after different prior user states. We also fit adjacent-turn regressions with prior-state $\times$ support-form interactions to test whether coupling depends jointly on the user's immediately preceding engagement state and the form of scaffolding provided. Standard errors for these adjacent-turn regressions were clustered by conversation. Coarser within-conversation before–after summaries around the first scaffolded assistant turn are reported as aggregate temporal diagnostics. They are used to contextualize the adjacent-turn claim rather than to estimate cumulative learning trajectories, because the first scaffolded turn may be selected by task difficulty, user motivation or an already developing exchange.

4.8 Robustness, privacy and LLM use

WildChat robustness analyses repeat the main patterns across user-facing model snapshots and coarse model families. These checks assess whether the headline patterns are reducible to a single WildChat model label. They are not interpreted as model-causal comparisons, because model labels are not experimentally assigned and may be confounded with time, user population, task mix and logging context.

The analyses are bounded by the observational design and by the nature of the labels. They support claims that everyday human–LLM conversations contain measurable learning-oriented engagement; that constructive engagement varies by intent, task ecology and depth; that assistant scaffolding is associated with richer constructive participation; and that support and engagement show adjacent-turn reciprocal ordering. They do not support claims that scaffolded support causes durable learning gains, that all constructive engagement reflects actual knowledge change or that the labelled conversational states perfectly measure learning itself.

LLMs were used as objects of study and as annotation tools. Production labels were generated with LLM-assisted workflows and then checked through parser rules, targeted update scripts and human validation. The authors also used LLM-based tools to assist with code drafting, debugging, analysis workflow organization and language editing. LLM-generated assistance was not used as a substitute for human verification of the validation samples or for interpretation of the statistical results.

All reported results are aggregate turn-level or conversation-level analyses. The study does not link conversations into user histories, estimate user-level longitudinal trajectories or publish raw conversation text; temporal analyses are confined to ordering within a conversation.

Data availability

This study analyses public human–LLM conversation corpora released by their original data providers: WildChat, LMSYS Chat-1M and ShareChat. Readers should obtain the raw public conversation data from the original releases and comply with the corresponding dataset licences, terms of use and redistribution restrictions. The manuscript does not redistribute raw message text, user identifiers or linked user histories.

Derived data sufficient to verify the reported aggregate figures, tables and statistical summaries are archived on Zenodo at https://doi.org/10.5281/zenodo.20995946. The same release is also available in the GitHub repository at https://github.com/CinderD/Informal-learning-in-everyday-human-LLM-interaction (tag `v0.1.0`, commit `31e10cfea930`). The release includes the numeric values underlying the main figures, de-identified conversation-level and adjacent-turn analytic labels without raw message text, table source files, confidence intervals, p values, false-discovery-rate q values, model-output summaries, provenance documentation and SHA-256 checksums. Any request for additional derived label files that are not included in the public release is assessed against the original dataset licences and privacy constraints.

Code availability

Custom code used to export source data and de-identified label tables, compute statistical analyses and generate figures and tables is archived with the derived-data release on Zenodo at https://doi.org/10.5281/zenodo.20995946 and is also available in the GitHub repository at https://github.com/CinderD/Informal-learning-in-everyday-human-LLM-interaction (tag `v0.1.0`, commit `31e10cfea930`). The repository excludes API keys, raw message text and files that cannot be redistributed under the original corpus terms. Reproducing the full annotation pipeline from raw conversations requires user-provided API credentials and access to the raw public corpora from their original sources.

Ethics statement

The study is a secondary analysis of public, naturalistic human–LLM conversation datasets. The authors did not recruit participants, intervene in conversations or attempt to identify users. Analyses are reported at aggregate turn or conversation level, and examples in the manuscript are paraphrased rather than copied from raw logs. Human validation was used to assess annotation quality on reviewed examples and did not involve interaction with the original dataset users.

Competing interests

H.L. and X.X. are employees of Microsoft Research Asia. The remaining authors declare no competing interests.

Author contributions

Author contributions are organized using the CRediT taxonomy. The contribution categories for this study are conceptualization, study design, data curation, annotation pipeline development, validation, statistical analysis, figure generation, writing of the original draft, manuscript review and editing, supervision and funding acquisition. Author initials and final role allocation: [to be added after the author list and order are confirmed]. All listed authors will approve the submitted version and will be accountable for their contributions.

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LLM use statement

Large language models were objects of study and were also used as annotation and writing-assistance tools. Production labels were generated with LLM-assisted workflows and checked through parser rules, targeted update scripts and human validation. The authors also used LLM-based tools to assist with code drafting, debugging, workflow organization and language editing. No LLM or AI tool is listed as an author, and the authors take responsibility for the final manuscript, analyses and interpretation. Additional details are provided in Methods.

Reporting summary and editorial policy checklist

A Nature Portfolio Reporting Summary and the journal-required editorial policy checklist have been prepared as separate submission files. The manuscript and supplementary information report the observational design, sampling and filtering rules, annotation workflow, human validation evidence, statistical model specifications, bootstrap procedures, multiple-comparison handling and data/code availability statements needed to assess reproducibility.

Figure source data

Source data for the numeric main figures are provided in `source_data/figure2_source_data.csv`, `source_data/figure3_source_data.csv`, `source_data/figure4_source_data.csv` and `source_data/figure5_source_data.csv`. Figure 1 is a conceptual framework figure and has no numeric source data. Supplementary table and regression source files are provided in `tables/` and `outputs/integrated_regression/`.

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Supplementary information

Supplementary information is provided below.

Appendix A Supporting methods

A.1 Corpus construction and analytic settings

The analyses use six task settings formed by crossing three public conversational corpora with two task ecologies: WildChat coding, WildChat writing, LMSYS coding, LMSYS writing, ShareChat coding and ShareChat writing. Coding and writing were selected because both are common everyday LLM use cases, yet they differ in the opportunities they create for visible learning-oriented engagement. Coding conversations often contain error messages, debugging attempts, execution failures and iterative repair. Writing conversations more often involve drafting, rewriting, shortening, tone adjustment and direct editing.

The analytic corpus contains 128,569 conversations and 981,470 total turns. WildChat contributes 31,878 coding conversations and 39,534 writing conversations; LMSYS contributes 32,114 coding conversations and 21,023 writing conversations; ShareChat contributes 2,481 coding conversations and 1,539 writing conversations. All reported analyses use the refreshed Level 1–4 outputs after the latest user-turn update pass and the subsequent LMSYS and ShareChat annotation runs.

Supplementary Table A1 summarizes how raw corpus records were converted into these analytic samples. It distinguishes source or pipeline metadata from analysis variables: coding/writing task ecology comes from LLM-assisted semantic filters, user framing comes from the conversation-level `learning_intent` annotation, and ShareChat strict-English status comes from a final conversation-level language flag applied after task filtering. Supplementary Table A2 records the retained provenance artifacts and script-hash prefixes used to audit the path from public source releases to the analytic samples and figure source data.

Table A1 Corpus filtering and classification pipeline. The table records the auditable steps used to construct the six main analytic settings. Task domain and user framing are analysis variables generated by the filtering and annotation pipeline, not native corpus metadata.

Corpus	Source records and first screen	Language and minimum-turn filter	Task classification source	Final analytic sample	User framing and audit trail
WildChat	Public WildChat-4.8M release. Raw conversations are obtained from the original release rather than redistributed with the manuscript. The retained analysis archive records the refreshed task-filter outputs, user-turn relabel/update logs and downstream source-data exports used for this manuscript version.	Retained task-filter outputs have at least four message turns. The task filters were English-oriented semantic filters; source language metadata were retained for analysis but were not used as an additional strict post-hoc exclusion in the main WildChat sample.	LLM-assisted semantic task filters labelled conversations as `coding_english` or `writing_english` with score threshold $\geq .70$ after keyword prefiltering. The retained filter outputs contain 31,879 coding conversations and 39,534 writing conversations; all retained records had task-filter scores above threshold.	31,878 coding; 39,534 writing.	User framing came from the conversation-level `learning_intent` annotation (`INTENTIONAL` versus `UNINTENTIONAL`), not from native metadata. The final WildChat user-turn update pass scanned 31,879 coding and 39,534 writing conversations; one coding conversation with missing `learning_intent` was omitted from Level 2 and downstream analyses.
LMSYS Chat	LMSYS Chat-1M source release: 1,000,000 conversations.	The conversion retained 777,453 English conversations and excluded 222,547 non-target-language conversations. Among retained English conversations, 236,372 had at least four turns.	The same LLM-assisted semantic task filters were run on the English pool with the minimum-turn screen applied inside the filter. The coding filter selected 32,114 coding conversations and the writing filter selected 21,023 writing conversations.	32,114 coding; 21,023 writing.	User framing came from the downstream `learning_intent` annotation. The conversion log also records redaction, moderation and model/source metadata where available; these fields were retained for robustness checks rather than used as task labels.
ShareChat	ShareChat Hugging Face release. The dataset card reports 142,808 public conversations and 660,293 public turns across five platforms. The local conversion starts from the five final-language-filtered CSV files, which contain 1,199,899 message rows and 129,584 unique platform-url conversations; role-normalization retained 1,196,969 rows mapped to `user` or `bot` (`user`=600,222; `bot`=596,747).	The first screen retained 69,915 conversations with at least four message turns. After task filtering, a strict-English screen retained only conversations with final conversation-level language flag `English`; conversations flagged non-English or mixed/non-English by this final flag were excluded.	The same LLM-assisted semantic filters selected 3,494 coding and 1,663 writing conversations before the strict-English screen. The strict-English screen retained 2,481 of 3,494 coding conversations and 1,539 of 1,663 writing conversations.	2,481 coding; 1,539 writing.	User framing came from the downstream `learning_intent` annotation. The strict-English summary recorded zero malformed JSON files; excluded counts were 1,013 coding and 124 writing conversations.

Note. A "conversation" is a role-ordered user–assistant record after corpus-specific conversion. The minimum-turn screen uses message count, not user–assistant round count. For ShareChat, the public dataset-card "turns" are not the same unit as the local message-row count used in conversion; our pipeline uses one row per role-labelled message and then counts user and assistant turns after role normalization. The term "strict-English" is reserved for the additional ShareChat final-language screen; WildChat and LMSYS Chat used English-oriented source or semantic filters as described above. Human validation was organized by task ecology rather than independently powered within each corpus; Supplementary Table C2 reports the domain-stratified agreement metrics.

Table A2 Corpus construction provenance artifacts. The table records the retained non-text provenance artifacts used to audit corpus construction. SHA-256 values are shown as 12-character prefixes for run scripts or manuscript-source scripts retained in the analysis archive. Raw conversation text remains governed by the original dataset releases and is not redistributed with the manuscript source.

Corpus	Source release and conversion unit	Selection ledger retained for this manuscript	Script/source-data provenance
WildChat	WildChat-4.8M public release; role-ordered user/assistant messages after task filtering.	Public release → English-oriented coding/writing semantic task filters with minimum four message turns → refreshed task-filter outputs (31,879 coding; 39,534 writing) → user-turn relabel/update pass → analytic sample (31,878 coding; 39,534 writing).	WildChat relabel/update run script SHA-256 prefix `c8eecef55858`; source-data exporter `206aaae44791`; camera-ready figure script `d9ecf7118f10`.
LMSYS Chat	LMSYS Chat-1M public release; one conversation record per public chat, converted to role-ordered messages.	1,000,000 source conversations → 777,453 English conversations → 236,372 English conversations with at least four turns → semantic task filters → analytic sample (32,114 coding; 21,023 writing).	Coding filter launcher `f54c079fda08`; writing filter launcher `be970385e72e`; after-filter annotation/analysis launcher `326e11709c41`; source-data exporter `206aaae44791`.
ShareChat	ShareChat Hugging Face release; public card reports 142,808 conversations and 660,293 public turns. Local conversion uses five final-language-filtered CSVs with one row per message and platform-url as the conversation key.	Five final-language-filtered CSVs (1,199,899 message rows; 129,584 platform-url conversations) → role-normalized user/bot rows (1,196,969 rows) → 69,915 conversations with at least four message turns → semantic task filters (3,494 coding; 1,663 writing) → strict-English screen → analytic sample (2,481 coding; 1,539 writing).	Coding filter launcher `f2ff9ca543b8`; writing filter launcher `2788818044e7`; after-filter annotation/analysis launcher `501e26e5a728`; source-data exporter `206aaae44791`.

For each conversation, the pipeline retained the ordered turn sequence, role labels, available topic and language metadata, learning-intent indicators and model metadata where available. Model metadata are used only for robustness checks where the corpus exposes usable model labels, because model assignment in naturalistic public conversations is observational and may reflect time, user population, task mix and logging context.

A.2 Annotation ontology

The codebook translates learning-science constructs into role-specific behavioural signatures. User turns are labelled for engagement. Passive engagement (`P`) marks acknowledgement, reception or minimal continuation without visible transformation. Active engagement (`A`) marks task-relevant action or follow-up, such as asking for a reformatted answer or applying supplied information. Constructive engagement (`C`) marks deeper work with ideas, such as testing an implication, revising a hypothesis, articulating a constraint, transferring a mechanism or extending an explanation. Emotional engagement is labelled separately because affective expressions were rare and not the main analytic outcome.

Assistant turns are labelled for scaffolding. Scaffolded support denotes assistant behaviour that structures, regulates, diagnoses or redirects the user's work while leaving some responsibility with the user. Non-scaffolded reference responses provide the requested information or deliverable without materially structuring the user's process. Scaffolded turns can also receive support-intent labels and support-means labels. Intention labels distinguish metacognitive support (`I1`), cognitive support (`I2`) and affective support (`I3`). Means labels distinguish feedback (`M1`), hinting (`M2`), instructing (`M3`), explaining (`M4`), modelling (`M5`) and questioning (`M6`). Means labels are non-mutually exclusive because a single assistant turn can both explain a mechanism and provide an example or hint. Supplementary Tables C16 and C17 report the support-intent and support-means profiles across the six task settings.

A.3 LLM-assisted annotation pipeline

The annotation pipeline combined prompt-based labelling with rule-based parser checks and targeted post-processing. During development, prompts were iterated against validation examples and disagreement cases. Common failure modes included treating brief acknowledgements as active engagement, treating any long assistant answer as scaffolding, collapsing directive instructions into scaffolding without checking whether responsibility remained with the user, and over-labelling sparse support means.

Production labels were generated at turn level. Parser checks enforced the expected label schema, normalized label variants and flagged missing or malformed outputs. Post-processing rules were used only when a recurring error pattern could be specified transparently, such as a narrow class of user-turn false positives. Prompt versions, run metadata, disagreement files and update outputs were retained with the analysis artifacts so that validation cases can be traced without reproducing raw conversation text in the manuscript.

Table A3 Filtering and classifier validation audit. The table separates corpus-construction filters from analysis labels and reports the strongest validation or audit evidence available in the current artifacts. Metrics from the writing-210 validation set are domain-level checks, whereas the user-framing audit deliberately sampled across corpus-by-task settings.

Variable or filter	How it was assigned	Validation or audit evidence	Interpretation boundary
Task ecology: coding versus writing	LLM-assisted semantic task filters selected conversations matching coding-oriented or writing-oriented use after minimum-turn screening. The filters were English-oriented and used a score threshold of 0.70 after keyword prefiltering where applicable.	The filtering ledger records retained counts and thresholds for each corpus (Supplementary Table A1). In the writing-210 validation artifacts, the conversation-topic field showed high human–LLM agreement (F1=0.973, MCC=0.872, Gwet AC1=0.936; support=210), but this is a domain-level check rather than a full cross-corpus task-filter audit.	Task ecology is treated as an analytic classification of conversation content, not native dataset metadata or randomized assignment.
User framing: intentional versus unintentional	Conversation-level LLM-assisted `learning_intent` annotation marked conversations explicitly oriented toward understanding, exploration or learning versus those without explicit learning-oriented framing.	A human-verified positive-oversampled audit reviewed 450 first user turns across all six corpus-by-task settings (75 per setting; 50 production-positive and 25 production-negative). Against the production labels, overall intentional-framing F1 was 0.857, MCC was 0.677 and Gwet AC1 was 0.671; task-stratified F1 was 0.924 for coding and 0.779 for writing.	User framing is a measured textual signal of explicit learning orientation, not a stable learner trait, motivation measure or causal exposure. Because the audit oversampled production-positive cases, it validates classification boundaries rather than estimating population prevalence.
Language filtering	WildChat and LMSYS Chat used English-oriented source or semantic filters. ShareChat received an additional strict-English screen after task filtering, retaining only records with final conversation-level language flag `English`.	The ShareChat strict-English audit recorded zero malformed JSON files and explicit exclusions after task filtering: 1,013 coding and 124 writing conversations were removed by the final language screen. In the writing-210 validation artifacts, language showed near-perfect agreement (F1=0.998, MCC=0.705, Gwet AC1=0.995; support=210), with MCC lower because nearly all reviewed records were English.	The term strict-English is used only for ShareChat's final post-task language screen. Mixed or non-English ShareChat records were excluded from the main analytic sample.
Turn-level labels: user engagement and scaffolded support	Production annotation used the final codebook, parser checks and post-processing rules. Constructive engagement and scaffolded support are the primary user-side outcome and assistant-side exposure.	Domain-stratified human–human codebook validation shows strong agreement for constructive engagement (coding F1=0.85; writing F1=0.89) and scaffolded support (coding F1=0.87; writing F1=0.97), with full per-label codebook metrics in Supplementary Table C2. Human-confirmed production audits across all six corpus-by-task settings showed acceptable or better reliability for user engagement and assistant labels: the 600-case user audit had constructive-vs-non-constructive F1=0.922, MCC=0.840 and Gwet AC1=0.837; the 180-case assistant audit had scaffolded-support F1=0.912, MCC=0.590 and Gwet AC1=0.791, with support-form F1 values ranging from 0.691 to 0.938 (Supplementary Table C3).	These labels support the main association and temporal analyses. The production audits are positive-oversampled and validate label boundaries rather than prevalence.

Production annotation used Azure OpenAI chat-completions deployments. The archived per-conversation `annotation_meta` fields record the schema, prompt version, annotation deployment and API version. Coding-oriented annotation used the V12 coding prompt `v21style_prompt_v5m`; WildChat coding outputs record deployment `gpt-5.1_2025-11-13` with API version `2024-10-21`, whereas LMSYS Chat and ShareChat coding outputs record deployment `gpt-5.1` with API version `2024-12-01-preview`. Writing-oriented annotation used the writing-topic prompt `v21style_prompt_v5m_writingtopic_r20260130_r60_intentlist_cp_cc_i3_refine_v10` and deployment `gpt-5.1-chat` with API version `2024-12-01-preview`. The final writing calls used JSON-object or JSON-schema response constraints, 60-second request timeouts, maximum completion budgets of 300 tokens for conversation-level labels and 350 tokens for turn-level labels, and did not pass an explicit temperature parameter because the deployment used the provider default. The final coding-from-filtered-conversation runner likewise did not pass an explicit temperature parameter. Where LLM-assisted semantic filtering was invoked, it used deployment `gpt-4o_2024-11-20` with API version `2024-10-21`; English-language checks used temperature 0.0 and semantic topic checks used temperature 0.1. Retry policies were specified in code: coding runs used up to five attempts, exponential backoff for rate limits and a fixed retry delay for other transient failures; writing runs used up to six attempts, disabled SDK-level retries, used exponential backoff with jitter for rate limits, timeouts and server errors, and fell back from JSON-schema response format to JSON-object format if unsupported. Content-filter blocks returned empty analyses and were logged rather than silently imputed.

Rule-based post-processing was part of the production annotation definition rather than a separate analysis choice. Coding outputs did not use a broad post-processing chain in the final filtered-conversation runner, but WildChat user turns underwent the targeted update pass described in Section A.4. Writing outputs used the best-pipeline chain retained in the run artifacts: assistant scaffolding recovery, assistant support-detail filling, strict user Active filtering, strict user Constructive filtering and strict Emotional filtering. Production manifests and logs retain UTC timestamps for auditable runs; verified production windows include WildChat coding from 15–22 February 2026, ShareChat coding/writing from 30 May–1 June 2026 and LMSYS Chat coding/writing from 31 May–24 June 2026. API keys and service endpoints are intentionally not reproduced in the manuscript.

A.4 Human verification and update provenance

Human verification was used to quantify label reliability and identify boundary cases. We use this evidence in three distinct ways. First, human–human agreement assesses whether the codebook was operational for trained reviewers; Supplementary Table C2 reports this codebook-agreement layer. Second, production-label checks compare final or post-processed LLM-assisted production labels with human review where a direct audit target was available; Supplementary Table C3 consolidates these human–LLM agreement checks. Third, post-update audits test whether targeted update passes reduced known residual false positives after prompt and parser development. Coding validation combined three review rounds, with 658 task turns and 643 turns after deduplication. Writing validation used 210 reviewed conversations with final human-reviewed labels. Because many labels are sparse or prevalence-imbalanced, agreement is reported with per-label F1, Matthews correlation coefficient and Gwet's AC1 rather than with a single global agreement statistic.

The final user-turn check contained 100 coding examples and 100 writing examples reviewed after the update pass. These examples were used to identify residual false positives and calibrate patch-style update filters, especially for Active and Passive user-turn labels. The last WildChat user-turn update pass scanned 31,878 coding conversations and 39,534 writing conversations, applying 1,815 coding updates and 21 writing updates. LMSYS Chat and ShareChat were then processed with the same refreshed annotation workflow, construct definitions and post-processing logic. The main manuscript reports the refreshed labels produced by these final production runs.

Because user framing carries an important contextual role, the consolidated production-label audit includes a human-verified check for the conversation-level learning-intent label. This audit sampled 450 first user turns, with 75 cases from each corpus-by-task setting and with production-positive cases deliberately oversampled. Overall intentional-framing agreement with the production label was F1=0.857, MCC=0.677 and Gwet AC1=0.671; task-stratified F1 was 0.924 for coding and 0.779 for writing. This audit validates the label boundary for user framing but is not used to estimate population prevalence because of the positive-oversampled design.

For user engagement production labels, the consolidated audit includes a 600-case check across the six corpus-by-task settings, with 100 user turns per setting and production constructive cases deliberately oversampled. The final labels were human-confirmed after item-level review. Against these human-confirmed labels, the production constructive-versus-non-constructive distinction had F1=0.922, MCC=0.840 and Gwet AC1=0.837. One-vs-rest metrics for the four user engagement levels were also high: F1=0.922 for constructive, 0.883 for active, 0.941 for passive and 0.930 for none (Supplementary Table C3). Because the sample is positive-oversampled, it validates label boundaries and production-label reliability rather than estimating population prevalence.

For assistant production labels, the consolidated audit includes a 180-case check across the six corpus-by-task settings, with 30 assistant turns per setting and scaffolded-support and support-form positives deliberately oversampled. The final labels were human-confirmed after item-level review. Broad scaffolded-support agreement was F1=0.912, MCC=0.590 and Gwet AC1=0.791. Support-form one-vs-rest F1 ranged from 0.691 to 0.938 across M1–M6 (Supplementary Table C3). Because the sample is positive-oversampled, it validates assistant-label boundaries rather than estimating support-form prevalence.

The validation evidence for turn-level engagement and support labels is therefore layered: human–human agreement supports codebook operability, and the targeted production audits directly check final user engagement and assistant scaffolding/support-form production labels across all six settings. Cross-corpus use of the labels is supported by applying the same schema, parser checks and post-processing rules to all three corpora and by reporting setting-level robustness checks in the Results and Supplementary Tables.

A.5 Outcome construction

At the turn level, the focal user-side outcome is a binary indicator for constructive engagement. At the conversation level, we use both the count of constructive user turns and an indicator for whether at least one constructive user turn occurs. Constructive ratios divide constructive user turns by total user turns; group-level constructive-ratio contrasts are turn-weighted ratios computed from total constructive user turns over total user turns within each group. Cognitive engagement ratios include passive, active and constructive cognitive engagement.

At the assistant side, scaffolding is operationalized as scaffolded support. Conversation-level scaffolding presence indicates whether a conversation contains at least one scaffolded assistant turn. Support-form analyses condition on conversations or turns with scaffolded support and then compare the presence versus absence of each means label. These comparisons are descriptive because means labels can co-occur and because conversations are not randomly assigned to receive a support form.

Behavioural depth is computed from turn counts and turn ordering. Post-answer depth counts turns after the first assistant response. Persistence after failure is analysed only for coding-oriented conversations because debugging failures, error messages and failed-execution contexts are more directly observable in coding than in writing.

A.6 Statistical model specification

The main analyses use simple estimands matched to the claim being made. Descriptive prevalence analyses report proportions within the six task settings. Context analyses compare user framing, task ecology and depth groups. Conversation-level support associations use Poisson generalized linear models for constructive-turn counts and logistic models for binary constructive-engagement outcomes. Adjacent-turn analyses use assistant-to-user and user-to-assistant turn pairs to estimate adjacent-turn conditional probabilities.

For the Section 2.3 adjusted conversation-level models, the Poisson outcome is the raw count of constructive user turns in a conversation, not a rate. The primary model does not include a user-turn offset. Conversation length is adjusted directly through total turns, so the exponentiated Poisson coefficient should be read as a multiplicative association with expected constructive-turn count conditional on observed conversation length and other covariates. The generic setting-specific mean model is $\mathbb{E}(C_i) = \exp(\alpha_s + \beta_s S_i + \gamma_s^\top X_i)$, where C_i is constructive-turn count, S_i is scaffolded-support presence and X_i contains the observed conversation-level context listed in Supplementary Table C6. The companion logistic model predicts whether a conversation contains any constructive user turn using scaffolded-support presence and the available setting-specific context. The user-framing stratified Poisson models use the same Poisson specification after stratifying by framing and omitting user framing from the covariate set.

Covariates are included when they correspond to observed compositional differences at the analysis level. Depending on the analysis, these include user framing, task indicators in pooled models, turn counts, emotional ratios, error-related variables and turn index. Percentage-point contrasts are reported as effect sizes. Confidence intervals and p values for these contrasts use 1,000 conversation-level bootstrap resamples; adjacent-turn contrasts use 1,000 conversation-cluster bootstrap resamples rather than individual turn-pair resampling to account for repeated pairs within conversations. The adjusted Poisson and logistic estimates in Fig. 3 use model-based standard errors. Poisson overdispersion was checked with the Pearson χ^2 statistic divided by residual degrees of freedom. Dispersion ranged from 1.16 to 2.06 across the six task settings; scaling the Poisson standard errors with a quasi-Poisson dispersion adjustment preserved positive scaffolded-support associations and statistical distinguishability in every setting. To check exposure specification, we also refitted the same setting-specific Poisson mean model with user turns represented as the exposure offset; Supplementary Table C9 reports quasi-Poisson scaled rate ratios. Negative-binomial models were not used as the primary specification; the sensitivity analyses focus on variance misspecification and exposure specification under the same mean model rather than changing the main estimand. Support-form bracket tests use Benjamini–Hochberg false-discovery-rate adjustment within each six-form comparison family.

The Section 2.2 constructive-rate sensitivity uses a grouped-binomial logistic model at the conversation level. The numerator is the number of constructive user turns and the denominator is total user turns in the conversation. The predictors match the any-constructive context model: user framing, task ecology, length bucket and dataset fixed effects. Standard errors use a conversation-robust sandwich estimator. This sensitivity is included because any-constructive-turn outcomes partly reflect the number of turns available for observing at least one constructive turn.

The integrated adjacent-turn regression combines three feature groups in one logistic model: user state before the assistant turn, assistant scaffolding features, and adjacent-turn context. The pooled specification includes dataset fixed effects. A model/source sensitivity uses exact WildChat model labels, LMSYS model names recovered from conversation identifiers and ShareChat assistant/source families recovered from conversation identifiers. These adjusted estimates are interpreted as measured-covariate-adjusted associations. They do not remove unobserved user motivation, prior expertise, task difficulty or time-varying conversational state.

A.7 Temporal scale

The temporal analyses focus on adjacent-turn ordering because it is the scale at which the logs most directly identify sequence. The adjacent-turn analyses ask whether the immediately following user turn is more likely to be constructive after scaffolded support than after a non-scaffolded reference response, whether that contrast depends on the prior user state and whether support-form signals differ across prior user states. The reverse adjacent-turn analysis asks whether the assistant is more likely to provide scaffolded support after constructive, active or passive user turns.

Coarser before–after summaries ask a different question: whether constructive engagement is higher after the first scaffolded assistant turn than before it. These are retained only as exploratory boundary checks, not as treatment-effect or accumulation estimates. The first scaffolded turn may occur precisely because the user is struggling, because a difficult task has emerged or because the exchange is already developing. The main temporal claim is therefore scale-specific: the clearest evidence concerns adjacent-turn, state- and form-conditioned coupling rather than a demonstrated global trajectory.

Appendix B Robustness and interpretive boundaries

B.1 WildChat model-snapshot robustness

WildChat includes user-facing model metadata that make it possible to repeat major descriptive patterns across model snapshots and coarse model families. These analyses preserved the main directional patterns reported in the main text: coding generally showed higher cognitive engagement than writing; intentionality gaps were visible across most user-facing models; scaffolded conversations were generally more constructive; and adjacent-turn contrasts were more interpretable than coarser within-conversation contrasts. Supplementary Figures D1 and D2 report six-setting temporal and framing diagnostics. Supplementary Figures D5–D9 are WildChat-only model-snapshot checks because comparable user-facing model-snapshot structure is not available across all three corpora.

These checks are used as robustness analyses rather than model-causal comparisons. WildChat model labels are not experimentally assigned. They may be confounded with calendar time, user population, task mix, interface changes and logging context. For this reason, the manuscript does not claim that a specific model family caused the observed engagement differences.

B.2 Cross-dataset and model/source regression checks

The Results claims were checked for cross-dataset consistency at the level appropriate to each estimand. Section 2.1 is a descriptive prevalence claim: cognitive engagement, constructive engagement and scaffolded support were observable in all six settings, with constructive engagement consistently the highest-level and less frequent user-side signal. Section 2.2 is primarily descriptive: user framing, task ecology and interaction depth organize where constructive engagement appears, with the same directional depth gradient across settings. As a simple model-based check, we fitted a conversation-level logistic regression for whether a conversation contained at least one constructive user turn, using user framing, task ecology, length bucket and dataset fixed effects (Supplementary Table C10). Because this any-constructive outcome is partly affected by the number of user turns available for observing at least one constructive turn, we also fitted a grouped-binomial constructive-rate sensitivity model, using constructive user-turn counts out of total user turns as the outcome and the same contextual predictors (Supplementary Table C11). Formal uncertainty estimates are otherwise reported where the claim involves a contrast or model coefficient. Section 2.3 uses conversation-level bootstrap confidence intervals and p values for turn-weighted scaffolded versus non-scaffolded constructive-ratio contrasts, adjusted Poisson/logistic model estimates and an offset-rate Poisson sensitivity using user turns as the exposure (Supplementary Table C9). Section 2.4 reports support-form confidence intervals and between-stratum FDR-adjusted q values in the main figure.

Section 2.5 reports conversation-cluster bootstrap confidence intervals for adjacent-turn lifts and cluster-robust regression estimates for the integrated adjacent-turn models.

The main percentage-point contrasts were directionally consistent across the six settings. Constructive-ratio contrasts comparing conversations with versus without scaffolded support were positive and statistically distinguishable from zero in all six settings, with effect sizes ranging from +0.99 to +7.34 percentage points. Adjacent-turn constructive lifts were also positive in all six settings, with five of six contrasts statistically distinguishable from zero; LMSYS writing was positive but weaker (+0.27 percentage points, 95% CI -0.01 to +0.56, $p = .061$). Supplementary Table C8 reports these effect sizes, confidence intervals and p values.

Integrated adjacent-turn regressions were used as robustness checks rather than as the primary narrative claim. These models predicted whether the next user turn was constructive after combining user-state features, assistant-scaffolding features and adjacent-turn context. Because model metadata differ by corpus, we report both within-setting models and pooled specifications. The within-setting models show the dataset-by-task pattern directly, with recoverable model/source fixed effects added separately inside each setting (Supplementary Table C12). The pooled specification includes dataset fixed effects; a sensitivity replaces these with exact WildChat model labels, LMSYS model names recovered from conversation identifiers and ShareChat assistant/source families recovered from conversation identifiers. This is interpreted as model/source adjustment rather than as a causal comparison among model versions. Model/source heterogeneity was detectable, but it did not absorb the scaffolding signal. Adding broad scaffolded support to user/context/model controls improved fit (LR $\chi^2_1 = 509.7$, $p < .001$). Because M1–M6 are support forms nested within scaffolded support rather than independent covariates, we interpret the decomposed models as asking which forms carry additional signal among scaffolded turns, not as a test of scaffolded support after removing its own components. Adding M1–M6 descriptors within scaffolded support improved fit (LR $\chi^2_6 = 829.8$, $p < .001$), and the full scaffolding block remained strongly informative after controls (LR $\chi^2_7 = 1339.6$, $p < .001$; Supplementary Table C14). Adding prior-state $\times$ support-form interactions also improved fit (pooled model/source fixed effects: LR $\chi^2_{18} = 297.3$, $p < .001$; Supplementary Table C15). Fully decomposed coefficients clarify the source of this form-level signal: prior user state is the largest predictor of immediate constructive follow-up, and M4 explaining is the most stable positive support-form coefficient (Supplementary Table C13).

B.3 Label uncertainty

The label scheme intentionally separates top-level constructs from finer support forms. Top-level labels such as constructive engagement and scaffolded support have clear behavioural anchors, while fine-grained support-form labels are interpreted as aggregate form-pattern signals rather than isolated item-level judgments. This is why the main claims rely most heavily on constructive engagement, scaffolded support presence, support-form patterns that are consistent across settings and temporal contrasts that can be expressed with simple conditional probabilities.

B.4 Selection into support

The most important threat to causal interpretation is selection into support. More complex tasks, more motivated users or already-developing exchanges may both elicit scaffolded support and produce constructive follow-up. Conversation-level adjusted models reduce measured compositional differences but cannot remove unobserved selection. Adjacent-turn analyses reduce the temporal window and condition on the immediately preceding user state, but they still cannot identify randomized exposure effects. The safest interpretation is therefore state-dependent coupling between assistant support and user engagement.

B.5 Privacy-preserving reporting

The study analyses naturalistic conversations, so the manuscript avoids reproducing raw conversation text. Operational examples are de-identified and paraphrased to illustrate decision rules without exposing user content. Aggregate statistics, model summaries and validation metrics are reported in the manuscript and supplementary tables. The analyses do not link conversations into user histories or estimate longitudinal user-level trajectories; temporal claims are restricted to within-conversation turn ordering.

Appendix C Supporting tables

Table C1 Operational examples for turn-level labels. Examples are de-identified and paraphrased from validation review patterns; they illustrate the decision rule rather than reproducing raw conversation text.

Construct	Paraphrased turn	Annotation rationale
`P` passive user engagement	"Thanks, that makes sense."	Minimal uptake or acknowledgement without new reasoning, evaluation or task transformation.
`A` active user engagement	"Can you convert that answer into a command I can run in my setup?"	Task-relevant follow-up or manipulation of supplied information, but without elaborating or testing an idea.
`C` constructive user engagement	"If the cache expires before the asynchronous callback returns, would the same race condition still happen?"	The user tests an implication, boundary condition or transfer of the prior explanation.
Scaffolding	"First isolate the failing step; that will show whether the issue is parsing or state management."	The assistant structures the user's process and leaves diagnostic or revision work with the user.
`M1` feedback	"Your diagnosis is close; the bug is in the index offset, not the loop boundary."	Evaluative feedback on the user's prior attempt or understanding.
`M2` hinting	"Check the parser log around the first malformed token before changing the model."	A cue or partial path that helps the user proceed without fully resolving the task.
`M3` instructing	"First split the condition into two branches, then add a regression test."	Explicit steps or strategy that organizes the user's work.
`M4` explaining	"The exception occurs because the event loop is already running, so a nested call cannot acquire it."	A rationale or mechanism that clarifies why the issue occurs.
`M5` modeling	"A small pattern you can adapt is: validate input, normalize it, then pass it to the writer."	A sample approach intended to be emulated or adapted rather than merely copied.
`M6` questioning	"Which constraint matters more here: preserving tone or shortening the paragraph?"	A question that advances problem solving or reflection.

Table C2 Domain-stratified human–human codebook agreement for labels used in the main analyses. Coding uses the latest three-round validation rerun over 658 task turns, 643 after deduplication; writing uses the final writing-210 validation labels. These validation sets assess whether human reviewers could apply the coding and writing codebook consistently; they are not, by themselves, direct estimates of production LLM-label accuracy and are not separately powered within each public corpus. F1, Matthews correlation coefficient (MCC) and Gwet's AC1 are reported per binary label. Coding MCC and AC1 are recomputed from the exported support and positive-count fields; writing metrics are from the final writing agreement report. Rare degenerate labels with no positive cases in a validation set are omitted. Production-label checks are reported in Supplementary Table C3, and post-update audits are described in Supplementary Section A.4.

Domain	Role	Label	F1	MCC	Gwet AC1	Support
Coding	User	Active (`A`)	0.75	0.72	0.87	315
Coding	User	Constructive (`C`)	0.85	0.82	0.90	315
Coding	User	Passive (`P`)	0.88	0.87	0.96	315
Coding	User	Emotional (`E`)	0.83	0.82	0.97	315
Coding	Assistant	Scaffolding	0.87	0.69	0.70	328
Coding	Assistant	Feedback (`M1`)	0.90	0.89	0.95	328
Coding	Assistant	Hinting (`M2`)	0.79	0.75	0.89	328
Coding	Assistant	Instructing (`M3`)	0.81	0.77	0.92	328
Coding	Assistant	Explaining (`M4`)	0.86	0.79	0.84	328
Coding	Assistant	Modeling (`M5`)	0.84	0.79	0.87	328
Coding	Assistant	Questioning (`M6`)	0.82	0.81	0.95	328
Writing	User	Active (`A`)	0.73	0.68	0.81	90
Writing	User	Constructive (`C`)	0.89	0.83	0.86	90
Writing	User	Passive (`P`)	0.80	0.81	0.99	90
Writing	User	Emotional (`E`)	0.80	0.79	0.98	90
Writing	Assistant	Scaffolding	0.97	0.80	0.93	120
Writing	Assistant	Feedback (`M1`)	0.89	0.80	0.80	120
Writing	Assistant	Hinting (`M2`)	0.82	0.70	0.71	120
Writing	Assistant	Instructing (`M3`)	0.71	0.67	0.86	120
Writing	Assistant	Explaining (`M4`)	0.84	0.70	0.70	120
Writing	Assistant	Modeling (`M5`)	0.94	0.91	0.95	120
Writing	Assistant	Questioning (`M6`)	0.96	0.96	0.99	120

Table C3 Consolidated human–LLM production-label agreement. User-framing audit sampled 450 first user turns across the six corpus-by-task settings, with production-positive intentional-framing cases deliberately oversampled. User engagement audit sampled 600 user turns, with 100 cases per setting and production constructive cases deliberately oversampled. Assistant audit sampled 180 assistant turns, with 30 cases per setting and scaffolded-support and support-form positives deliberately oversampled. Final human labels were confirmed after item-level review. Metrics compare final LLM-assisted production labels with human-confirmed labels using one-vs-rest binary coding for each label. Because the samples are positive-oversampled, these metrics validate label boundaries rather than estimating population prevalence.

Role	Label	Audit N	Prod. +	Human +	Precision	Recall	F1	MCC	Gwet AC1	Accuracy
Conversation-level framing										
Conversation	Intentional framing	450	300	239	0.770	0.967	0.857	0.677	0.671	0.829
User engagement										
User	Constructive (`C`)	600	300	327	0.963	0.884	0.922	0.840	0.837	0.912
User	Active (`A`)	600	120	111	0.850	0.919	0.883	0.856	0.935	0.958
User	Passive (`P`)	600	90	80	0.889	1.000	0.941	0.934	0.978	0.983
User	None	600	90	82	0.889	0.976	0.930	0.920	0.974	0.980
Assistant scaffolding and support forms										
Assistant	Scaffolding	180	144	139	0.896	0.928	0.912	0.590	0.791	0.861
Assistant	Feedback (`M1`)	180	45	51	1.000	0.882	0.938	0.918	0.945	0.967
Assistant	Hinting (`M2`)	180	43	45	0.930	0.889	0.909	0.880	0.930	0.956
Assistant	Instructing (`M3`)	180	51	55	0.745	0.691	0.717	0.600	0.715	0.833
Assistant	Explaining (`M4`)	180	87	110	1.000	0.791	0.883	0.772	0.747	0.872
Assistant	Modelling (`M5`)	180	48	79	0.958	0.582	0.724	0.631	0.642	0.806
Assistant	Questioning (`M6`)	180	35	20	0.543	0.950	0.691	0.675	0.873	0.906

Table C4 Analytic model families used in the main analyses. The table summarizes descriptive and conversation-level model families. Adjusted models are interpreted as covariate-adjusted associations in naturalistic data, not as causal treatment-effect estimates.

Analysis family	Unit and outcome	Main contrast or model	Scope
Engagement prevalence	User turns and conversations; cognitive, constructive and emotional engagement indicators.	Descriptive proportions by task setting, task ecology and user framing.	Establishes where learning-oriented engagement is visible; no causal contrast is implied.
Engagement composition	Cognitively engaged user turns; passive, active and constructive depth levels.	Composition of cognitive engagement within each task setting.	Motivates constructive engagement as the focal high-level outcome while preserving the broader cognitive-engagement hierarchy.
Interaction depth	Conversations; at least one constructive user turn.	Probability of constructive engagement across behavioural-depth buckets.	Depth is interpreted as interactional opportunity and task development, not as a manipulated exposure.
Scaffolding presence	Conversations; constructive-turn count and any constructive turn.	Setting-specific Poisson GLM for constructive-turn count with scaffolded-support presence and observed conversation-level context; no user-turn offset in the primary count model. Setting-specific logistic model for whether any constructive user turn occurs. Estimable covariates are listed in Supplementary Table C6.	Adjusted estimates reduce measured compositional differences but remain associational because support is selected into by task difficulty and user state. Pearson overdispersion was checked and quasi-Poisson scaling preserved the direction of the Poisson estimates.

Table C5 Additional analytic model families and robustness checks. These analyses qualify the main associations by support form, temporal scale and available model or source metadata.

Analysis family	Unit and outcome	Main contrast or model	Scope
Support-form associations	Scaffolded conversations; constructive engagement ratios.	Mean differences for conversations containing a given support-means label versus scaffolded conversations without that label, summarized by user framing and task ecology.	Descriptive because means labels are non-exclusive and can co-occur in the same assistant turn.
Local temporal coupling	Adjacent assistant–user or user–assistant turn pairs; next constructive user turn or next scaffolded assistant turn.	Conditional probability contrasts around adjacent turns; detailed estimands are listed in Supplementary Table C7.	Interpreted as local ordering and state-dependent coupling, not as randomized support effects.
Integrated adjacent-turn regression	Adjacent assistant–user turn pairs; next constructive user turn.	Logistic regression combining prior user state, scaffolded-support presence, support means, task/framing context, prior-state $\times$ support-form interactions and dataset or model/source fixed effects.	Tests whether the local coupling pattern depends jointly on user state and support form; standard errors are clustered by conversation.
Coarse temporal boundary checks	Conversations containing scaffolded support; constructive engagement before and after the first scaffolded turn.	Mean after-minus-before contrast reported as an exploratory appendix check.	Not used as evidence of accumulation because the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.
WildChat model robustness	WildChat conversations or turns; main prevalence, association and temporal outcomes.	Repetition across user-facing model snapshots or coarse model families.	Robustness summaries only; model labels are observational and may be confounded with time, user mix and task mix.

Table C6 Setting-specific covariate sets for the Section 2.3 adjusted models. Each model was fitted separately within the six task settings. Task ecology is therefore fixed within a setting and is not interpreted as a within-setting coefficient. The table lists the estimable covariates used to adjust the scaffolded-support association; pooled context models use task ecology separately as shown in Supplementary Tables C10 and C11.

Settings	Primary Poisson count model	Logistic any-constructive model	Sensitivities and notes
WC coding; LMSYS coding; SC coding; WC writing; LMSYS writing; SC writing	Scaffolded-support presence; user framing; total turns; emotional-expression ratio; observable error marker; persistence-after-failure marker; high-persistence marker.	Scaffolded-support presence; user framing; total turns.	The offset-rate sensitivity used the same Poisson mean model but represented user turns with an exposure offset. User-framing stratified Poisson models omitted user framing because it was fixed within stratum. Error and persistence markers are most interpretable in coding-oriented contexts but are retained as observed conversation-level context when present.

Table C7 Temporal estimands for adjacent-turn support–engagement coupling. Main temporal analyses use adjacent turn pairs. Coarser before–after summaries are exploratory boundary checks because scaffolded turns are selected by task difficulty, user motivation and conversation trajectory.

Estimand	Definition	Reported in main text	Scope
Overall adjacent-turn lift	Next-turn constructive probability after scaffolded support versus a non-scaffolded reference response, estimated within each task setting.	Figure 5a	Local association between support type and immediately following constructive engagement; not a randomized support effect.
Prior-state adjacent contrast	Same next-turn contrast, stratified by prior constructive, active and passive user states.	Figure 5b	Tests whether local coupling depends on the user's immediately preceding state.
Reverse state-conditioned support	Probability that the next assistant turn is scaffolded after constructive, active and passive user turns.	Figure 5c	Describes how scaffolding supply varies with the user's current engagement state.
Prior-state support-form regression	Adjacent-turn logistic regression for next-turn constructive engagement, adding prior-state $\times$ M1–M6 interactions.	Figure 5d and Supplementary Table C15	Tests whether local coupling depends jointly on prior user state and support form.
Around-first-scaffold contrast	Mean constructive engagement after versus before the first scaffolded assistant turn, among conversations containing scaffolded support.	Supplementary Figure D1	Exploratory boundary check only; the first scaffolded turn may be selected by difficulty or an already developing exchange.

Table C8 Uncertainty estimates for key support–engagement contrasts. Constructive-ratio contrasts are turn-weighted conversation-bootstrap estimates, with ratios computed as total constructive user turns divided by total user turns within scaffolded versus non-scaffolded conversation groups. Post-answer depth contrasts use conversation-level bootstrap resampling. Adjacent-turn contrasts bootstrap conversations over assistant-to-user pairs. Percentage-point and turn differences are the effect sizes reported in the main figures.

Setting	Contrast	Effect	95% CI	p value
WC coding	Scaffolded − reference constructive ratio	+4.56 pp	[+4.15, +4.99]	$< .001$
WC coding	Scaffolded − reference post-answer depth	+2.23 turns	[+2.11, +2.35]	$< .001$
LMSYS coding	Scaffolded − reference constructive ratio	+2.50 pp	[+2.19, +2.83]	$< .001$
LMSYS coding	Scaffolded − reference post-answer depth	+1.42 turns	[+1.34, +1.52]	$< .001$
SC coding	Scaffolded − reference constructive ratio	+7.34 pp	[+5.29, +9.27]	$< .001$
SC coding	Scaffolded − reference post-answer depth	+3.36 turns	[+2.69, +4.01]	$< .001$
WC writing	Scaffolded − reference constructive ratio	+0.99 pp	[+0.82, +1.16]	$< .001$
WC writing	Scaffolded − reference post-answer depth	+3.13 turns	[+2.96, +3.30]	$< .001$
LMSYS writing	Scaffolded − reference constructive ratio	+1.09 pp	[+0.87, +1.31]	$< .001$
LMSYS writing	Scaffolded − reference post-answer depth	+2.10 turns	[+1.91, +2.28]	$< .001$
SC writing	Scaffolded − reference constructive ratio	+3.20 pp	[+1.55, +4.89]	$< .001$
SC writing	Scaffolded − reference post-answer depth	+3.38 turns	[+1.92, +4.64]	$< .001$
WC coding	Adjacent scaffolded − reference lift	+2.68 pp	[+2.22, +3.09]	$< .001$
LMSYS coding	Adjacent scaffolded − reference lift	+2.01 pp	[+1.54, +2.50]	$< .001$
SC coding	Adjacent scaffolded − reference lift	+6.21 pp	[+4.35, +8.10]	$< .001$
WC writing	Adjacent scaffolded − reference lift	+1.13 pp	[+0.88, +1.41]	$< .001$
LMSYS writing	Adjacent scaffolded − reference lift	+0.27 pp	[-0.01, +0.56]	.061
SC writing	Adjacent scaffolded − reference lift	+3.35 pp	[+0.80, +6.62]	.025

Table C9 Offset-rate sensitivity for scaffolded-support Poisson models. The primary Fig. 3b Poisson model uses raw constructive-turn counts with conversation length entered as a covariate. As a rate sensitivity, this table refits the same setting-specific mean model with `offset(log(user_turns))`. Cells report the scaffolded-support rate ratio with 95% confidence intervals and two-sided p values using quasi-Poisson scaled standard errors.

Setting	Offset rate ratio [95% CI], p value	Pearson dispersion	Conversations
WC coding	1.57 [1.47, 1.68], $< .001$	1.29	31,878
LMSYS coding	1.47 [1.38, 1.56], $< .001$	1.13	32,114
SC coding	1.58 [1.31, 1.91], $< .001$	1.29	2,481
WC writing	1.36 [1.26, 1.46], $< .001$	1.15	39,534
LMSYS writing	1.70 [1.50, 1.94], $< .001$	1.12	21,023
SC writing	1.65 [1.27, 2.15], $< .001$	1.32	1,539

Table C10 Conversation-level context model for constructive engagement. The outcome is whether a conversation contains at least one constructive user turn. The logistic regression includes user framing, task ecology, conversation-length bucket and dataset fixed effects. Cells report odds ratios with 95% confidence intervals and two-sided p values from conversation-robust standard errors. The reference category is unintentional framing, writing-oriented task, 2–3 user turns and WildChat.

Predictor	OR [95% CI], p value
Intentional framing	3.03 [2.93, 3.13], $< .001$
Coding task ecology	1.89 [1.71, 2.08], $< .001$
4–6 user turns	2.07 [1.99, 2.15], $< .001$
7+ user turns	3.58 [3.42, 3.74], $< .001$
LMSYS Chat	0.70 [0.68, 0.73], $< .001$
ShareChat	2.29 [2.12, 2.46], $< .001$
Conversations	128,569

Table C11 Conversation-level constructive-rate sensitivity model. The outcome is the number of constructive user turns out of total user turns in each conversation, modelled with a grouped-binomial logistic regression. The model includes user framing, task ecology, conversation-length bucket and dataset fixed effects, using user-turn counts as denominators to reduce the mechanical opportunity effect in any-constructive-turn outcomes. Cells report odds ratios with 95% confidence intervals and two-sided p values from conversation-robust standard errors. The reference category is unintentional framing, writing-oriented task, 2–3 user turns and WildChat.

Predictor	OR [95% CI], p value
Intentional framing	2.78 [2.68, 2.88], $< .001$
Coding task ecology	1.66 [1.47, 1.88], $< .001$
4–6 user turns	1.13 [1.09, 1.18], $< .001$
7+ user turns	1.06 [1.01, 1.11], .010
LMSYS Chat	0.72 [0.69, 0.75], $< .001$
ShareChat	2.30 [2.13, 2.50], $< .001$
Conversations	128,569
User turns	491,685
Constructive user turns	24,227

Table C12 Setting-level integrated adjacent-turn models. Each column reports a separate within-setting logistic regression for whether the next user turn is constructive. The scaffolded-support row comes from a broad scaffolded-support model. Support-form rows come from a decomposed model that includes broad scaffolded-support presence and co-occurring M1–M6 forms, so form coefficients compare variation within scaffolded support rather than replacing the broad scaffolded-support contrast. All models include prior user state, intentional framing, assistant-turn index and recoverable model/source fixed effects where available. Cells report odds ratios with 95% confidence intervals and two-sided p values from conversation-cluster robust standard errors.

Predictor	WC coding	LMSYS coding	SC coding	WC writing	LMSYS writing	SC writing
A2U pairs	92,042	71,910	9,001	124,214	55,580	5,825
Conversations	31,809	31,606	2,474	39,455	20,936	1,537
Model/source FE count	13	23	4	13	23	3
Scaffolded support	1.13 [1.08, 1.19] $< .001$	1.25 [1.16, 1.34] $< .001$	1.24 [1.09, 1.42] .001	1.57 [1.38, 1.78] $< .001$	1.04 [0.86, 1.26] .679	1.34 [0.98, 1.85] .068
Prior user constructive	5.20 [4.84, 5.59] $< .001$	6.01 [5.43, 6.66] $< .001$	4.13 [3.46, 4.93] $< .001$	8.28 [6.73, 10.19] $< .001$	11.89 [9.10, 15.54] $< .001$	6.23 [4.43, 8.76] $< .001$
Prior user active	1.67 [1.57, 1.76] $< .001$	1.58 [1.46, 1.71] $< .001$	1.56 [1.32, 1.83] $< .001$	1.91 [1.68, 2.18] $< .001$	2.18 [1.80, 2.64] $< .001$	1.47 [1.07, 2.01] .017
Prior user passive	2.19 [1.65, 2.92] $< .001$	2.01 [1.38, 2.92] $< .001$	1.82 [1.25, 2.66] .002	2.49 [1.69, 3.65] $< .001$	2.32 [1.29, 4.17] .005	0.86 [0.40, 1.84] .703
Intentional framing	1.57 [1.49, 1.66] $< .001$	1.46 [1.35, 1.58] $< .001$	1.57 [1.34, 1.83] $< .001$	1.02 [0.89, 1.16] .798	0.86 [0.70, 1.06] .150	1.37 [0.93, 2.02] .107
M1 feedback	1.19 [1.07, 1.31] .001	1.21 [0.98, 1.50] .079	1.28 [1.03, 1.60] .025	0.69 [0.53, 0.91] .008	0.81 [0.51, 1.31] .397	1.74 [1.12, 2.71] .013
M2 hinting	1.04 [0.95, 1.14] .429	0.90 [0.72, 1.11] .332	0.85 [0.65, 1.10] .209	1.52 [1.20, 1.93] $< .001$	1.63 [1.09, 2.44] .018	0.76 [0.45, 1.29] .314
M3 instructing	0.89 [0.83, 0.96] .003	0.84 [0.69, 1.04] .103	0.93 [0.77, 1.13] .491	1.36 [1.04, 1.78] .027	1.06 [0.71, 1.58] .779	0.94 [0.45, 1.94] .860
M4 explaining	1.21 [1.09, 1.35] $< .001$	1.36 [1.08, 1.72] .008	1.77 [1.35, 2.33] $< .001$	1.81 [1.40, 2.34] $< .001$	1.12 [0.75, 1.68] .570	1.09 [0.58, 2.06] .789
M5 modelling	0.86 [0.80, 0.93] $< .001$	0.73 [0.63, 0.86] $< .001$	0.76 [0.60, 0.96] .021	0.69 [0.47, 1.01] .055	1.04 [0.64, 1.68] .872	1.57 [0.94, 2.61] .084
M6 questioning	0.73 [0.62, 0.86] $< .001$	0.30 [0.21, 0.44] $< .001$	0.70 [0.54, 0.90] .006	1.13 [0.74, 1.72] .567	0.66 [0.36, 1.21] .179	1.66 [0.90, 3.08] .106

Table C13 Integrated adjacent-turn regression combining user state, assistant scaffolding and model controls. The outcome is whether the next user turn is constructive. The scaffolded-support row is estimated in a broad scaffolded-support model; M1, M4 and M6 rows are estimated in a decomposed support-form model that includes broad scaffolded-support presence and co-occurring support forms. Estimates are odds ratios with 95% confidence intervals and two-sided p values from conversation-cluster robust standard errors. The primary pooled model uses dataset fixed effects; the model/source sensitivity adds recoverable assistant model or source-family fixed effects.

Predictor	Pooled dataset FE	Pooled model/source FE	WildChat model FE
Scaffolded support	1.50 [1.44, 1.55], $< .001$	1.47 [1.41, 1.52], $< .001$	1.56 [1.49, 1.63], $< .001$
Prior user constructive	8.90 [8.45, 9.38], $< .001$	8.55 [8.11, 9.01], $< .001$	9.11 [8.53, 9.74], $< .001$
Prior user active	2.62 [2.52, 2.72], $< .001$	2.57 [2.47, 2.67], $< .001$	2.85 [2.71, 3.00], $< .001$
Prior user passive	2.08 [1.76, 2.44], $< .001$	2.07 [1.76, 2.44], $< .001$	2.32 [1.85, 2.90], $< .001$
Intentional framing	1.40 [1.35, 1.46], $< .001$	1.42 [1.37, 1.48], $< .001$	1.40 [1.33, 1.47], $< .001$
Coding task	1.37 [1.22, 1.54], $< .001$	1.37 [1.22, 1.54], $< .001$	1.21 [1.05, 1.40], .009
M1 feedback	0.79 [0.73, 0.86], $< .001$	0.80 [0.74, 0.86], $< .001$	0.74 [0.67, 0.82], $< .001$
M4 explaining	2.15 [1.99, 2.32], $< .001$	2.12 [1.96, 2.30], $< .001$	2.11 [1.92, 2.31], $< .001$
M6 questioning	0.82 [0.72, 0.93], .002	0.84 [0.74, 0.96], .008	1.08 [0.93, 1.25], .340
Model/source block	Reference	LR $\chi^2_{41} = 714.4$, $p < .001$	LR $\chi^2_{13} = 553.2$, $p < .001$

Table C14 Incremental contribution of scaffolding features in integrated adjacent-turn models. Likelihood-ratio block tests compare nested logistic regressions for whether the next user turn is constructive. All models include prior user state, user framing, task ecology, assistant-turn index and the indicated dataset or model/source fixed effects.

Scope	Added feature block	LR χ^2 (df)	p value
six task settings, dataset FE	Broad scaffolded support after user/context controls	584.6 (1)	$< .001$
six task settings, dataset FE	M1–M6 descriptors within scaffolded support	896.7 (6)	$< .001$
six task settings, dataset FE	Full scaffolding block after user/context controls	1481.3 (7)	$< .001$
six task settings, model/source FE	Broad scaffolded support after user/context controls	509.7 (1)	$< .001$
six task settings, model/source FE	M1–M6 descriptors within scaffolded support	829.8 (6)	$< .001$
six task settings, model/source FE	Full scaffolding block after user/context controls	1339.6 (7)	$< .001$
WildChat only, model FE	Broad scaffolded support after user/context controls	418.6 (1)	$< .001$
WildChat only, model FE	M1–M6 descriptors within scaffolded support	519.4 (6)	$< .001$
WildChat only, model FE	Full scaffolding block after user/context controls	937.9 (7)	$< .001$

Table C15 Prior user state and all six support forms jointly structure adjacent-turn constructive engagement. The first panel reports likelihood-ratio tests adding prior-state × M1–M6 interactions to the integrated adjacent-turn logistic model. The second panel reports pooled model/source fixed-effect estimates for each support form within each prior user state. Estimates are odds ratios with 95% confidence intervals and two-sided p values from conversation-cluster robust standard errors.

Panel	Scope / prior state	Term or test	Estimate
Interaction block	six task settings, dataset FE	Prior state × M1–M6	LR $\chi^2_{18} = 305.4$, $< .001$
Interaction block	six task settings, model/source FE	Prior state × M1–M6	LR $\chi^2_{18} = 297.3$, $< .001$
Interaction block	WildChat only, model FE	Prior state × M1–M6	LR $\chi^2_{18} = 186.0$, $< .001$
State-stratified model/source FE	prior constructive	M1 feedback	0.72 [0.64, 0.81], $< .001$
	prior constructive	M2 hinting	0.91 [0.79, 1.04], .155
	prior constructive	M3 instructing	0.92 [0.81, 1.05], .236
	prior constructive	M4 explaining	1.72 [1.45, 2.04], $< .001$
	prior constructive	M5 modelling	0.81 [0.70, 0.93], .003
	prior constructive	M6 questioning	0.95 [0.72, 1.25], .694
	prior active	M1 feedback	0.83 [0.72, 0.95], .008
	prior active	M2 hinting	1.14 [1.03, 1.27], .016
	prior active	M3 instructing	0.98 [0.89, 1.08], .722
	prior active	M4 explaining	2.13 [1.88, 2.41], $< .001$
	prior active	M5 modelling	0.83 [0.76, 0.91], $< .001$
	prior active	M6 questioning	0.75 [0.61, 0.93], .009
	prior passive	M1 feedback	2.16 [1.16, 4.04], .015
	prior passive	M2 hinting	1.79 [0.95, 3.37], .071
	prior passive	M3 instructing	1.52 [0.67, 3.43], .312
	prior passive	M4 explaining	2.88 [1.55, 5.35], $< .001$
	prior passive	M5 modelling	0.46 [0.20, 1.04], .061
	prior passive	M6 questioning	2.19 [1.12, 4.27], .021

Table C16 Support-intent labels are dominated by cognitive support. Scaffolded assistant turns can receive metacognitive (`I1`), cognitive (`I2`) or affective (`I3`) intent labels. Scaffolded-turn prevalence reports the share of scaffolded assistant turns containing each intent label. Constructive ratios compare scaffolded conversations containing at least one assistant turn with the intent label with scaffolded conversations without that intent label, using turn-weighted constructive user-turn ratios. Intent labels can co-occur, and affective intent is sparse, so these contrasts are interpreted descriptively.

Scope	Intent label	Scaffolded turns (%)	With intent (%)	Without intent (%)	Diff. (pp)	Conv. with / without
Pooled	`I1` metacognitive	8.7	5.0	6.9	-2.0	10,924 / 55,892
Pooled	`I2` cognitive	91.6	6.8	2.8	+4.0	61,942 / 4,874
Pooled	`I3` affective	0.8	12.4	6.4	+6.0	1,052 / 65,764
Coding	`I1` metacognitive	8.2	7.0	9.6	-2.6	6,782 / 36,179
Coding	`I2` cognitive	92.3	9.4	3.9	+5.6	40,279 / 2,682
Coding	`I3` affective	1.0	16.1	8.9	+7.2	778 / 42,183
Writing	`I1` metacognitive	9.6	2.5	2.8	-0.3	4,142 / 19,713
Writing	`I2` cognitive	90.4	2.8	1.8	+0.9	21,663 / 2,192
Writing	`I3` affective	0.6	3.9	2.7	+1.2	274 / 23,581

Table C17 Support-supply profile within scaffolded assistant turns. Values show the percentage of scaffolded assistant turns containing each scaffolding-means label. Support-form labels are not mutually exclusive, so rows do not sum to 100%. Coding-oriented scaffolded turns are dominated by explanation-heavy support, whereas writing-oriented scaffolded turns show a more mixed support profile.

Setting	M1 Feedback	M2 Hinting	M3 Instructing	M4 Explaining	M5 Modelling	M6 Questioning
WildChat coding	8.8	14.7	29.7	84.0	23.9	6.9
LMSYS coding	4.4	7.6	9.7	76.7	16.5	13.9
ShareChat coding	13.3	12.7	25.7	81.2	18.4	18.6
WildChat writing	16.6	9.6	59.2	29.5	15.0	6.9
LMSYS writing	11.3	10.9	52.3	26.9	10.3	13.4
ShareChat writing	24.2	23.2	22.9	45.2	23.5	38.1

Appendix D Supporting figures

The supporting figures in Appendix D separate two roles. Supplementary Figures D1 and D2 extend the main analyses across the six WildChat, LMSYS Chat and ShareChat task settings. Supplementary Figure D3 reports the support-form supply profile that contextualizes the main support-form associations, and Supplementary Figure D4 provides paraphrased support-form cards for the six scaffolded-support forms. Supplementary Figures D5–D9 are WildChat-only robustness displays because WildChat exposes user-facing model-snapshot metadata and a large enough robustness subset for these additional checks. LMSYS Chat and ShareChat do not provide directly comparable model-snapshot structure in the current analysis artifacts. The model-snapshot figures are descriptive robustness checks because model labels in naturalistic corpora are observational rather than experimentally assigned.

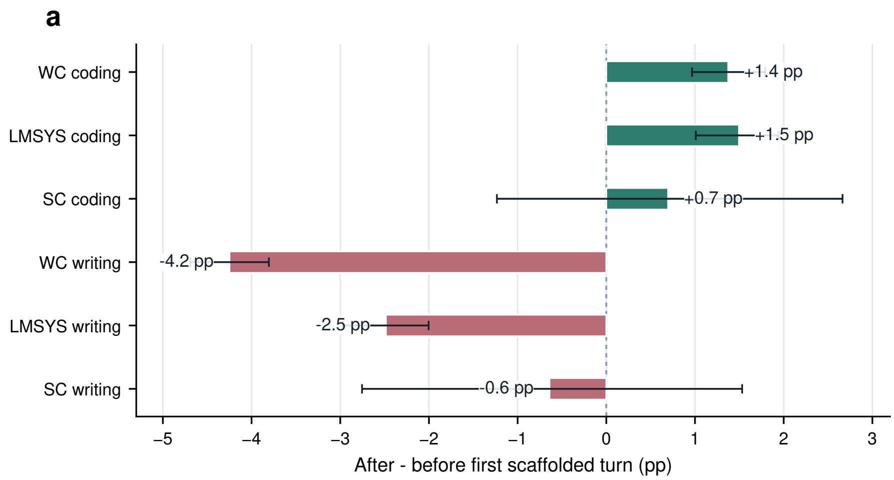

Fig. D1 Coarse around-first-scaffold contrasts are exploratory boundary checks across the six task settings. Bars show the difference in constructive engagement after versus before the first scaffolded assistant turn within conversations containing scaffolded support. Error bars show 95% conversation-bootstrap confidence intervals. These before–after contrasts are not used as treatment-effect or accumulation estimates because the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.

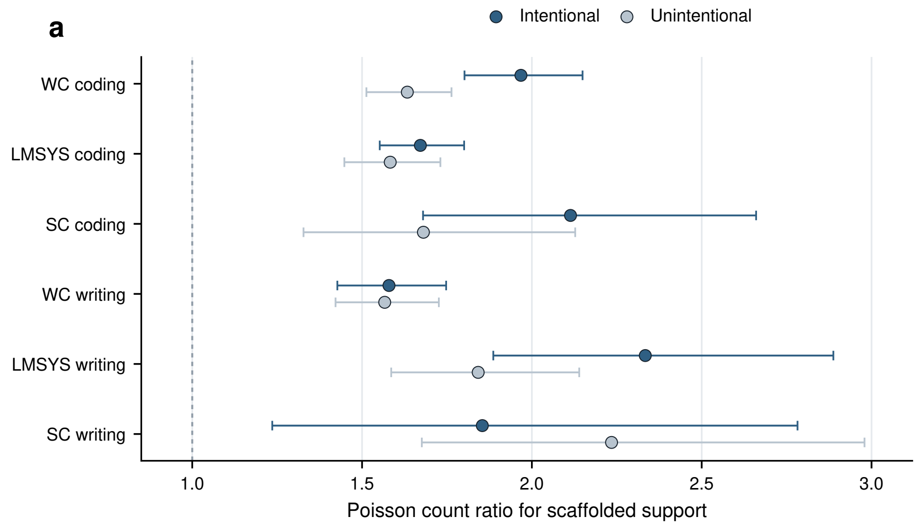

Fig. D2 Scaffolded-support associations by user framing across the six task settings. Stratified Poisson count-ratio estimates compare the association between scaffolded-support presence and constructive participation in intentional versus unintentional conversations. Error bars show 95% model-based confidence intervals. The figure is interpreted as descriptive moderation rather than evidence that framing causally changes support effectiveness, because user framing, task complexity and persistence are not randomly assigned.

Support-form supply within scaffolded assistant turns (%)

	M1	M2	M3	M4	M5	M6
WC coding	8.8	14.7	29.7	84.0	23.9	6.9
LMSYS coding	4.4	7.6	9.7	76.7	16.5	13.9
SC coding	13.3	12.7	25.7	81.2	18.4	18.6
WC writing	16.6	9.6	59.2	29.5	15.0	6.9
LMSYS writing	11.3	10.9	52.3	26.9	10.3	13.4
SC writing	24.2	23.2	22.9	45.2	23.5	38.1

WC, WildChat; LMSYS, LMSYS Chat; SC, ShareChat. Support-form labels are non-exclusive.

Fig. D3 Support-form supply within scaffolded assistant turns across the six task settings. Cells report the percentage of scaffolded assistant turns containing each non-exclusive support-form label. Rows do not sum to 100% because a single scaffolded turn can contain multiple support forms. The figure provides task-context background for the support-form association contrasts in Fig. 4.

M1 feedback
Evaluates a user's prior attempt

"Your diagnosis is close: the race is not in the callback itself, but in when the cache key is reused. Check that key before changing the loop."

Role: Returns the user's attempt as something to repair, revise or extend.

M2 hinting
Offers a partial cue

"Before rewriting the parser, inspect the first malformed token and compare it with the token immediately before it."

Role: Leaves solution work open while directing attention to a useful diagnostic path.

M3 instructing
Provides procedural steps

"First isolate the failing branch, then add a regression test, and only then refactor the duplicated condition."

Role: Organizes the user's next actions through explicit sequencing or strategy.

M4 explaining
Makes a mechanism visible

"The exception occurs because the event loop is already running; a nested call asks for a resource that cannot be acquired twice."

Role: Provides a rationale that can be tested, adapted or transferred.

M5 modelling
Shows an adaptable pattern

"A reusable structure is: validate input, normalize fields, then pass the cleaned object to the writer. You can adapt this skeleton to your case."

Role: Supplies an example or pattern intended for adaptation rather than only completion.

M6 questioning
Elicits a constraint or reflection

"Which constraint matters more here: preserving the author's tone or shortening the paragraph? That choice should determine the revision strategy."

Role: Prompts the user to specify priorities, constraints or next reasoning steps.

Fig. D4 Representative support-form cards for the six scaffolded-support forms. Cards show de-identified, paraphrased assistant replies that illustrate the annotation decision rule for each support-form label rather than reproducing raw conversation text. Support-form labels are non-exclusive, so a single assistant turn can receive more than one form label.

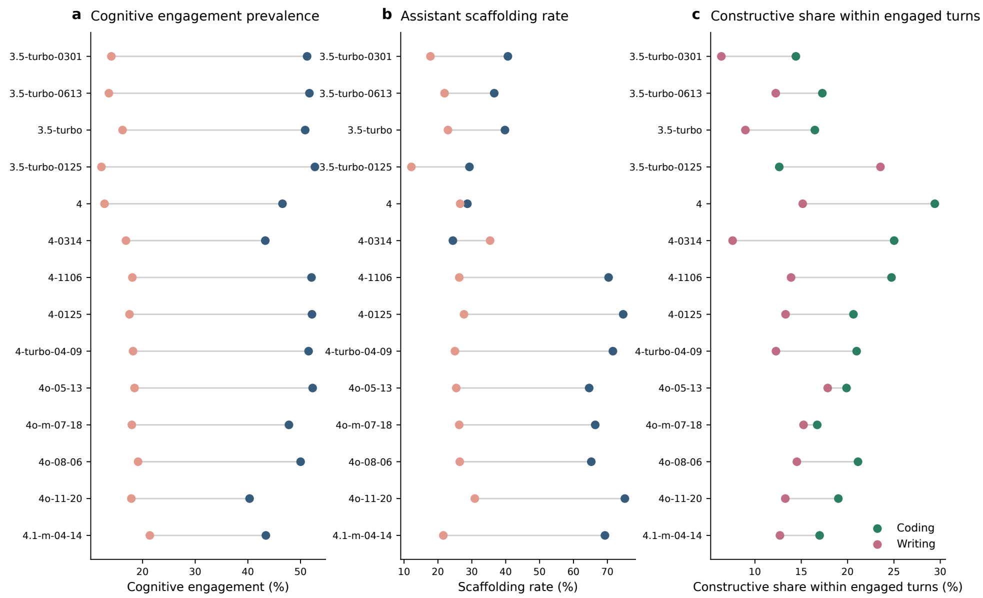

Fig. D5 WildChat model-snapshot prevalence profiles preserve the main domain contrast. Across user-facing WildChat model snapshots, coding-oriented conversations generally show higher cognitive engagement and higher assistant scaffolding rates than writing-oriented conversations. The model labels are observational and may reflect time, user mix and traffic composition, so the figure is used as a robustness check rather than a model-causal comparison.

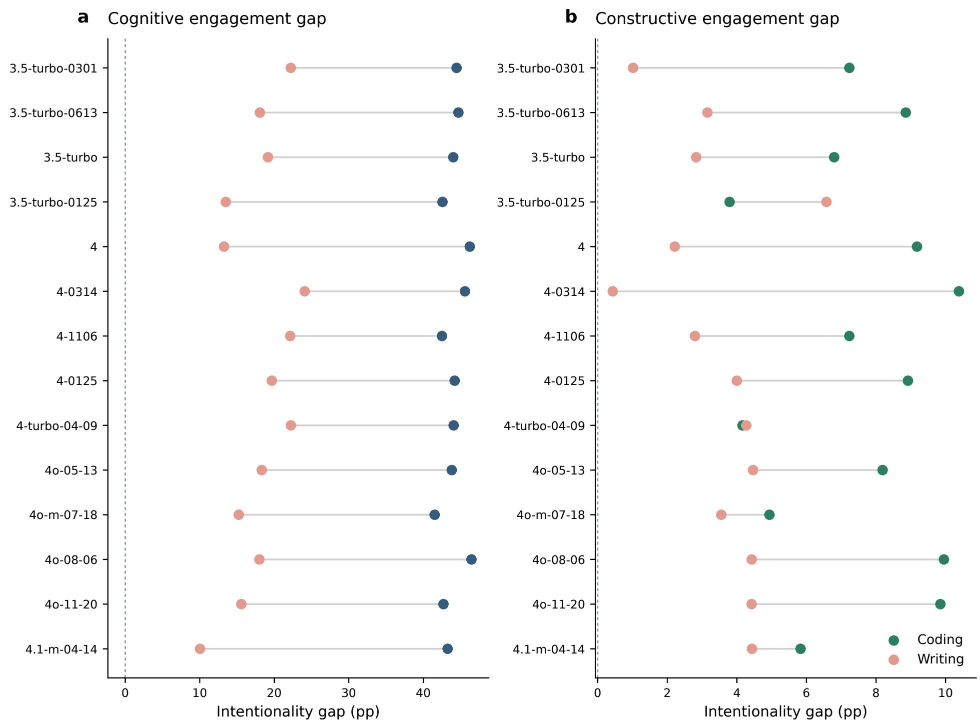

Fig. D6 Intentionality gaps are visible across WildChat model snapshots. The figure reports model-specific differences between intentional and unintentional conversations in cognitive and constructive engagement, separately for coding and writing. Most model snapshots preserve the directional pattern that intentionally framed conversations contain richer learning-oriented participation.

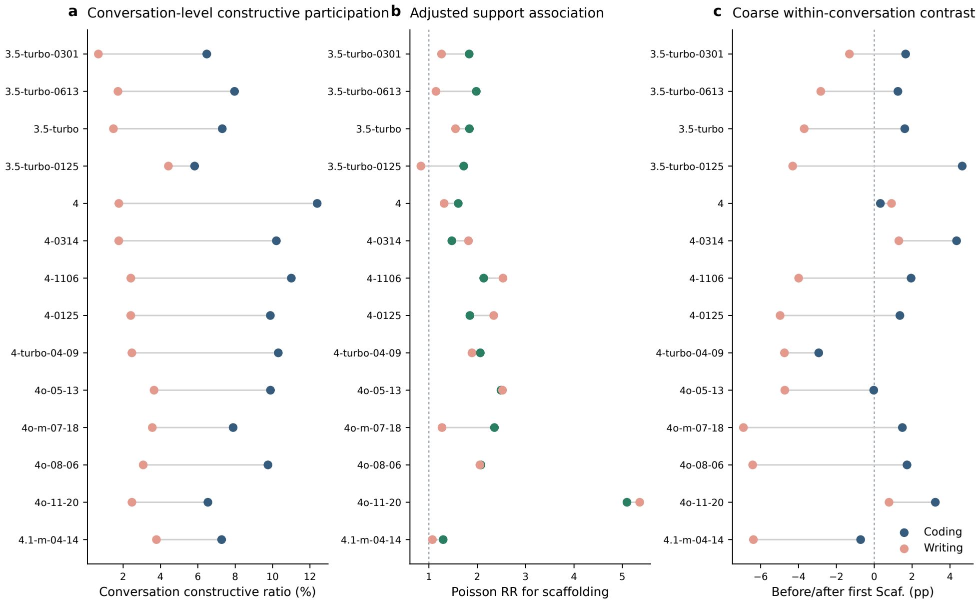

Fig. D7 Conversation-level support associations are broadly positive across WildChat model snapshots. The figure repeats major conversation-level analyses by user-facing WildChat model snapshot, including constructive participation, adjusted support associations and exploratory coarse before–after contrasts around the first scaffolded assistant turn. The pattern supports the distinction between stable conversation-level associations and adjacent-turn temporal claims.

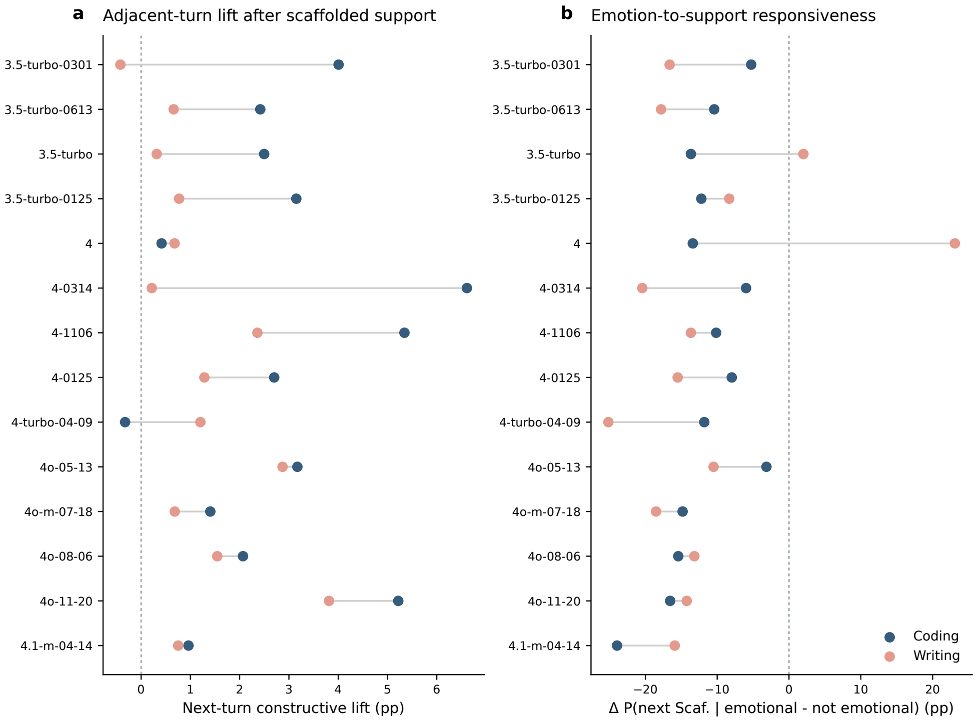

Fig. D8 Local temporal signals vary across WildChat model snapshots. The figure reports adjacent-turn constructive lifts and emotion-to-support responsiveness by model snapshot. Adjacent-turn constructive lifts are often positive, especially in coding-oriented conversations, but the magnitude varies across snapshots; emotion-to-support responsiveness is more heterogeneous and is treated as exploratory.

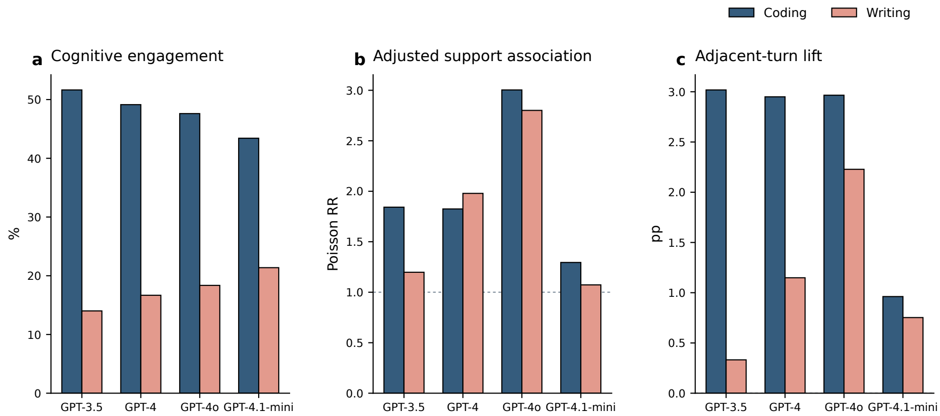

Fig. D9 Coarse model-family aggregation preserves the main directional patterns. WildChat model snapshots are grouped into coarse model families to summarize cognitive engagement prevalence, adjusted support associations and adjacent-turn constructive lift. Coding remains higher than writing across families, and the support-association and adjacent-turn lift patterns remain directionally consistent on average.